Burpsuite

Web application Pentesting

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Abstract

In order to protect online assets, web application security testing is an essential element of safeguarding them. Burp Suite has been a leader in this area for many years and it's still being used by safety professionals as well as Ethical hackers.

The primary role of Burp Suite is to assist security professionals, such as penetration testers and ethical hackers, in identifying and addressing vulnerabilities in web applications, which will be the subject of this report.

Disclaimer: This report is provided for educational and informational purpose only (Penetration Testing). Penetration Testing refers to legal intrusion tests that aim to identify vulnerabilities and improve cybersecurity, rather than for malicious purposes.

Introduction

In the first section of this report, we'll explore a helpful extension called "Autorize," which makes the process of testing authentication and authorization in web application security easier.

Moving on to the second part, we'll take a look at another useful extension known as "Burp Logger++," which assists many web experts in identifying issues on websites.

Burpsuite for Pentester: Autorize

Autorize = Authenticate + Authorize

Authorization includes any method by which a system grants or revokes permission to access specific data or actions. Meanwhile, Authentication is a process by which an individual or system authenticates themselves as being who they claim to be.

- Common vulnerabilities detected by Autorize
- Understanding the Functionality
- Installation and Setup
- Navigation and Configuration options
- Practical Demonstration of Autorize in Action

Common vulnerabilities detected by Autorize

It is primarily focused on identifying authorization-related vulnerabilities. It can help to identify some of the main types of vulnerabilities, such as:

- **Inadequate Role-Based Access Control (RBAC)**: It can uncover issues where user roles or permissions are not properly enforced, allowing users to access functionality or data they shouldn't have access to.
- **Broken Access Controls**: It can identify instances where access controls are not correctly implemented, leading to unauthorized access to resources or actions.
- **Insecure Direct Object References (IDOR)**: It can find situations where attackers can manipulate input to access other users' data or perform actions, they shouldn't be able to.
- **Forced Browsing**: It can help identify cases where an attacker can navigate directly to restricted areas of the application by manipulating URLs.

- **Insufficient Authorization**: It may detect situations where user roles or permissions are not properly enforced, allowing unauthorized actions to be performed.
- **Horizontal and Vertical Privilege Escalation**: It can find vulnerabilities that enable attackers to escalate their privileges within the application, either by impersonating other users or gaining additional permissions.
- **Business Logic Flaws**: Autorize may discover business logic vulnerabilities, where application workflows can be manipulated in unintended ways, potentially leading to unauthorized actions or data exposure.

Remember that the effectiveness of Autorize depends on how well it is configured and your tests are carried out.

Understanding the working of Autorize

Let's understand how Autorize works. Suppose, for instance, a web application implements user-based roles and supports cookie-based authentication.

Normal User: has access to general functionality but is not allowed to access admin functions and database (read-only access).

Admin User: has access to all functionality (read/write access).

Capture the normal user cookies and add them to Authorize. Re-log in with the Admin user access all the admin functionality and update some data to the database.

What will Autorize be doing now? Autorize is capturing all requests and changing the administrator cookie with your normal user's cookies when you are browsing an application, then sending them to server. See the server response, if the server behaves in the same way as legitimate Admin (like 200 OK in response) and no errors have been detected. The request was highlighted as a Red Bypass! Another request shows as a Green Enforced!.

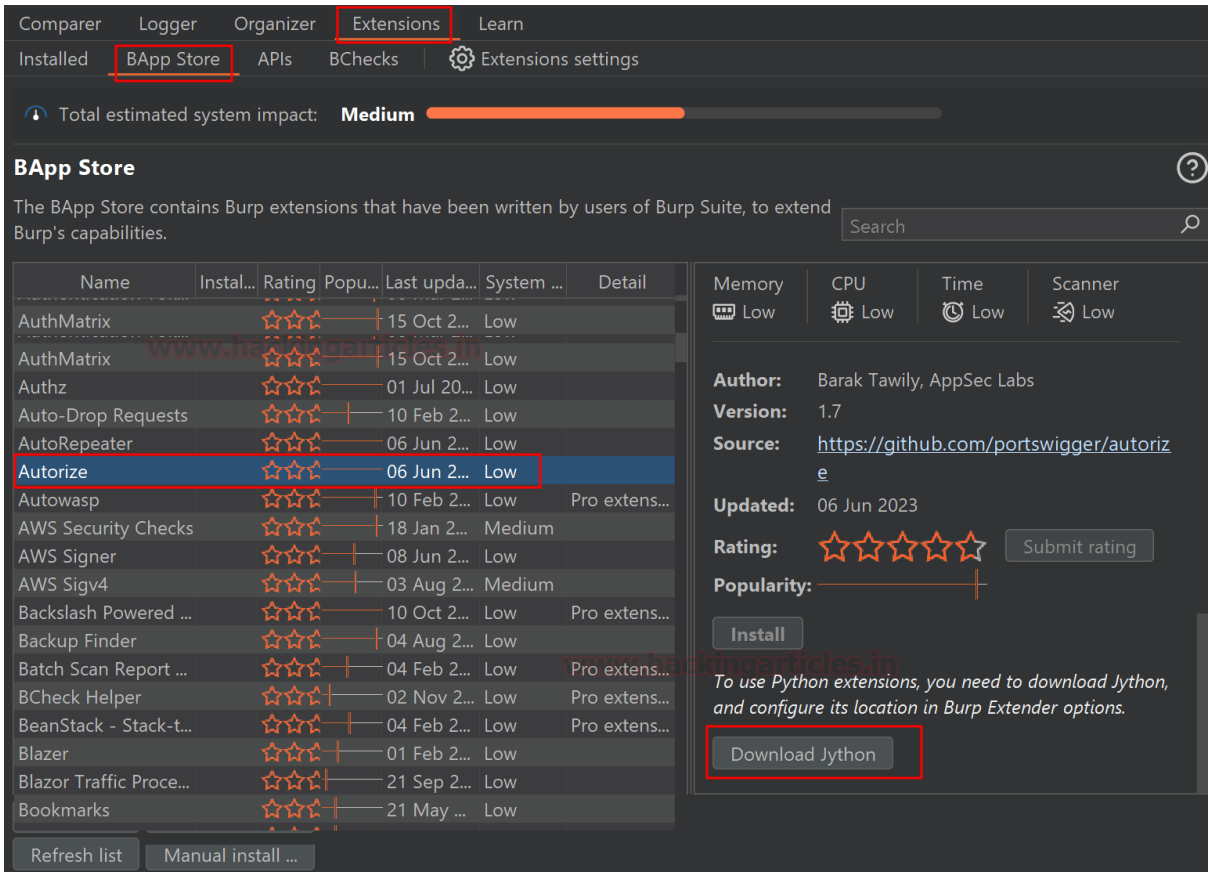
For every request sent to the server from a client, it will perform an automated test. With a large application, with over 30+ dynamic webpages, it's going to ease our work. There are a lot of URLs you need to test manually, so Autorize will do it for you.

Similarly, Autorize also detects an API endpoint problem in the same way. The authentication method must be checked for the API. Let's say an API uses a JWT token, you can control that by modifying its authorization header and identifying the authentication bypass issues with the APIs.

Installation and Setup

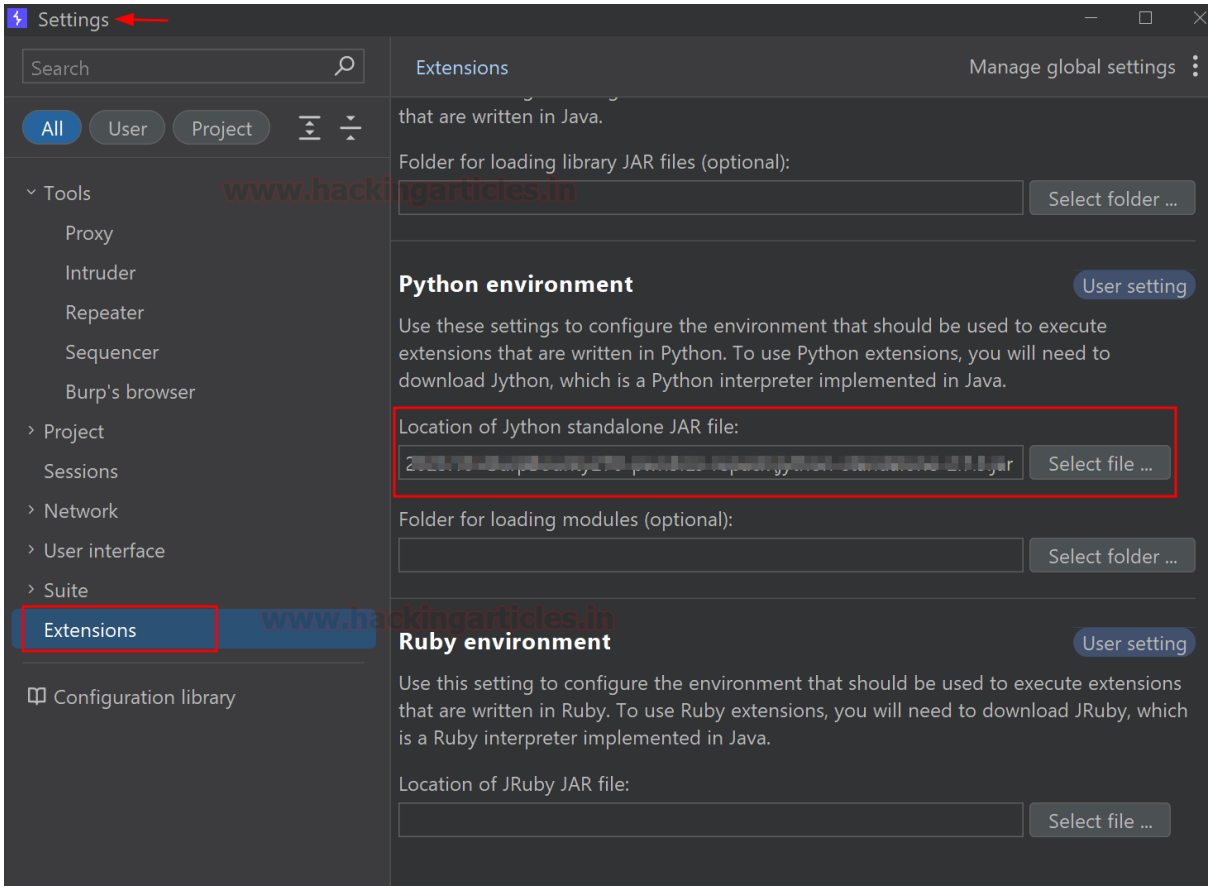
From the Bapp Store, you can download and install the extension. Select Bapp Store in Extensions. You can search for 'Authorize', or you can just look down. Click on it, scroll down to the right side.

The extension is built in Python, you will see that 'Jython' needs to be installed first.

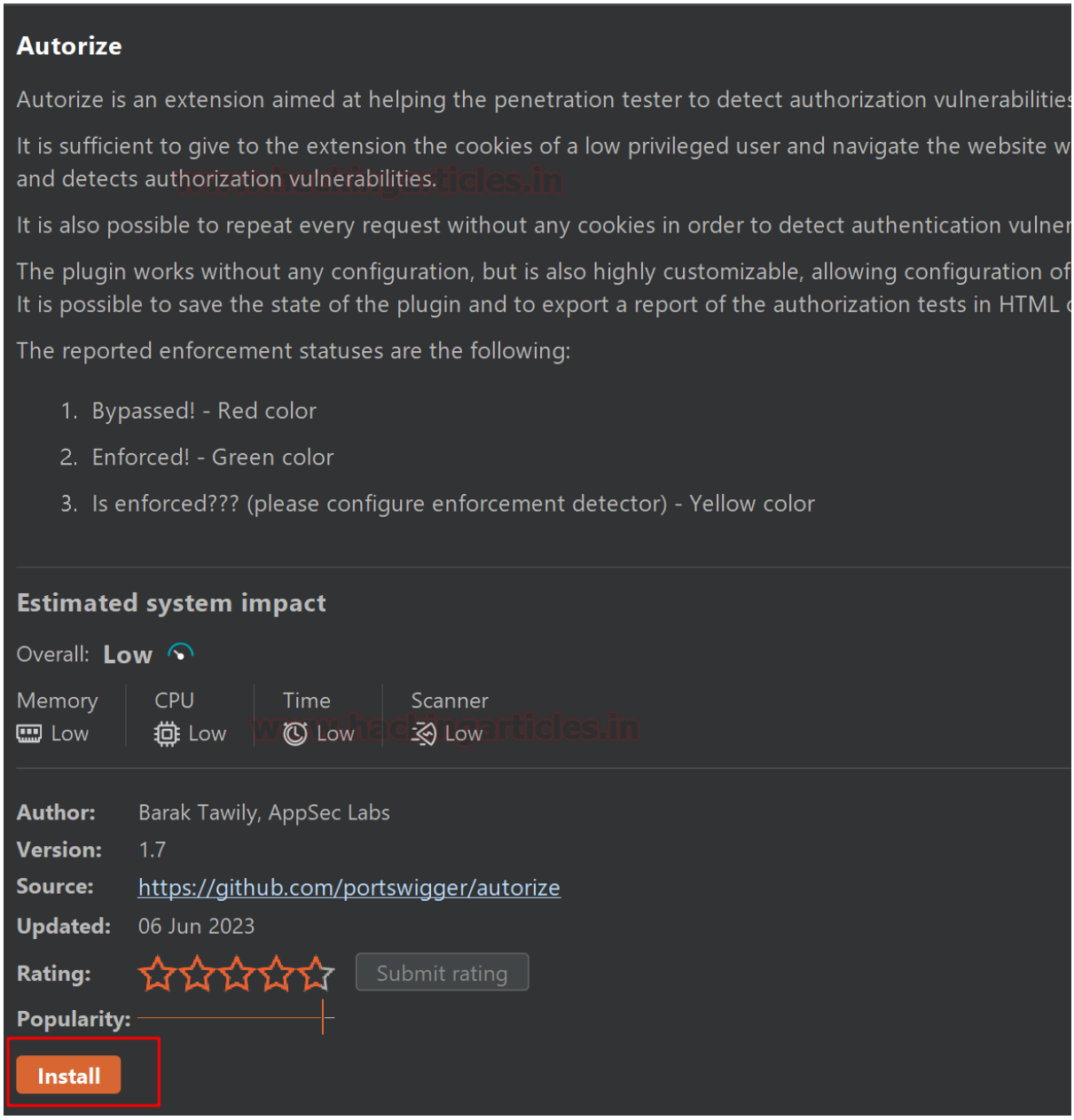


Browse the below link and download 'Jython Standalone'.

After downloading go to Setting > Extension > on the right side under Python Environment browser the Jython file. This environment has been successfully set up for Jython.



Restart the Burp program and follow this path to install Authorize on BApp Store. You'll notice that the install button is highlighted. You can click on it and install it.



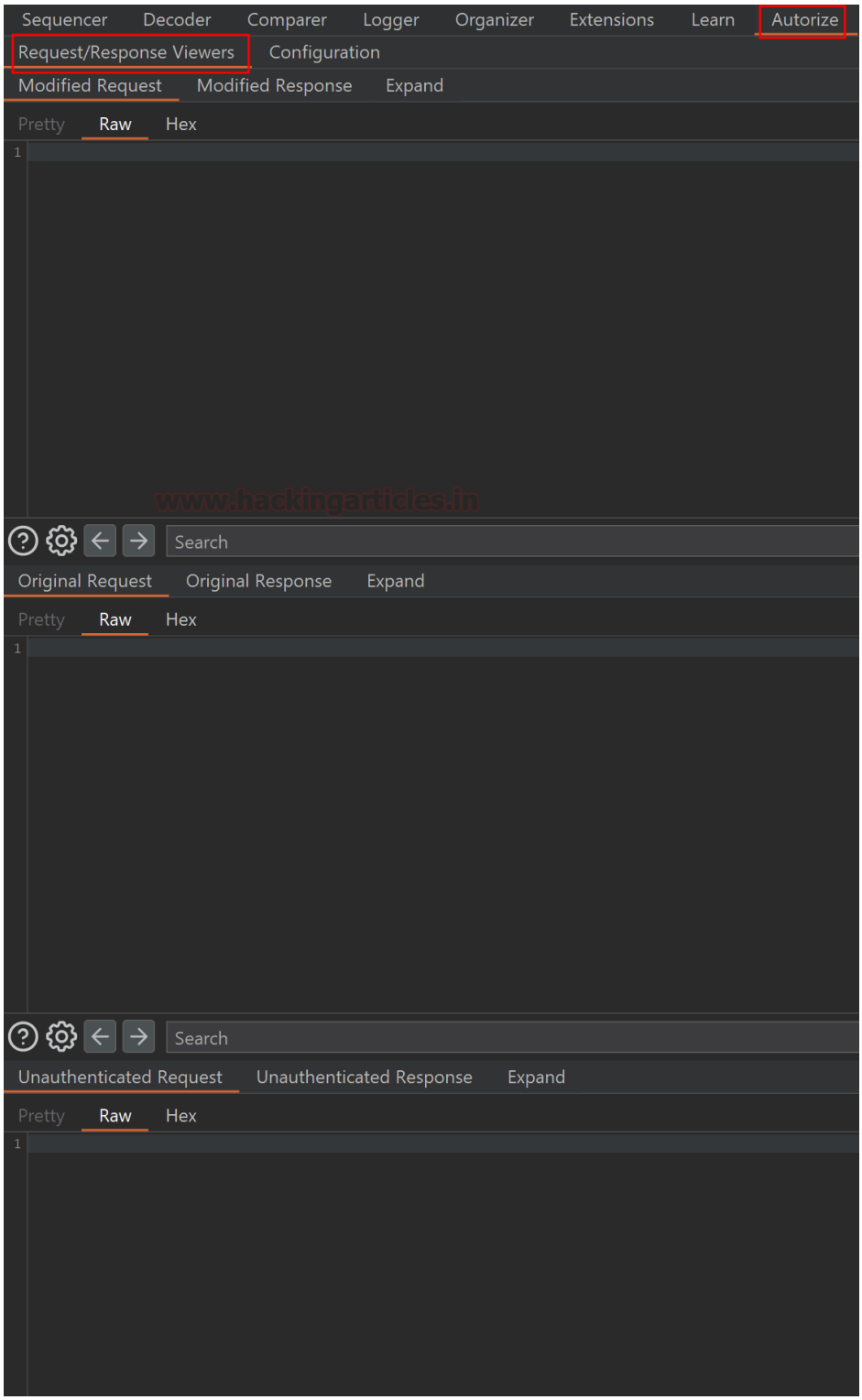
The Authorize tab will appear in the bar after successful installation.

Navigating and Configuration Options

There are two tabs under the Authorize section, the first one is Request/Response Viewers tab and the other one Configuration tab.

Request/Response Viewers: The Request/Response tab will display complete information about the particular request you capture within Authorize and choose. The manipulated

request will be displayed under the Modified Request section, the Original Request tab will display the original/unmodified request, and the Unauthenticated request will display the unauth request.

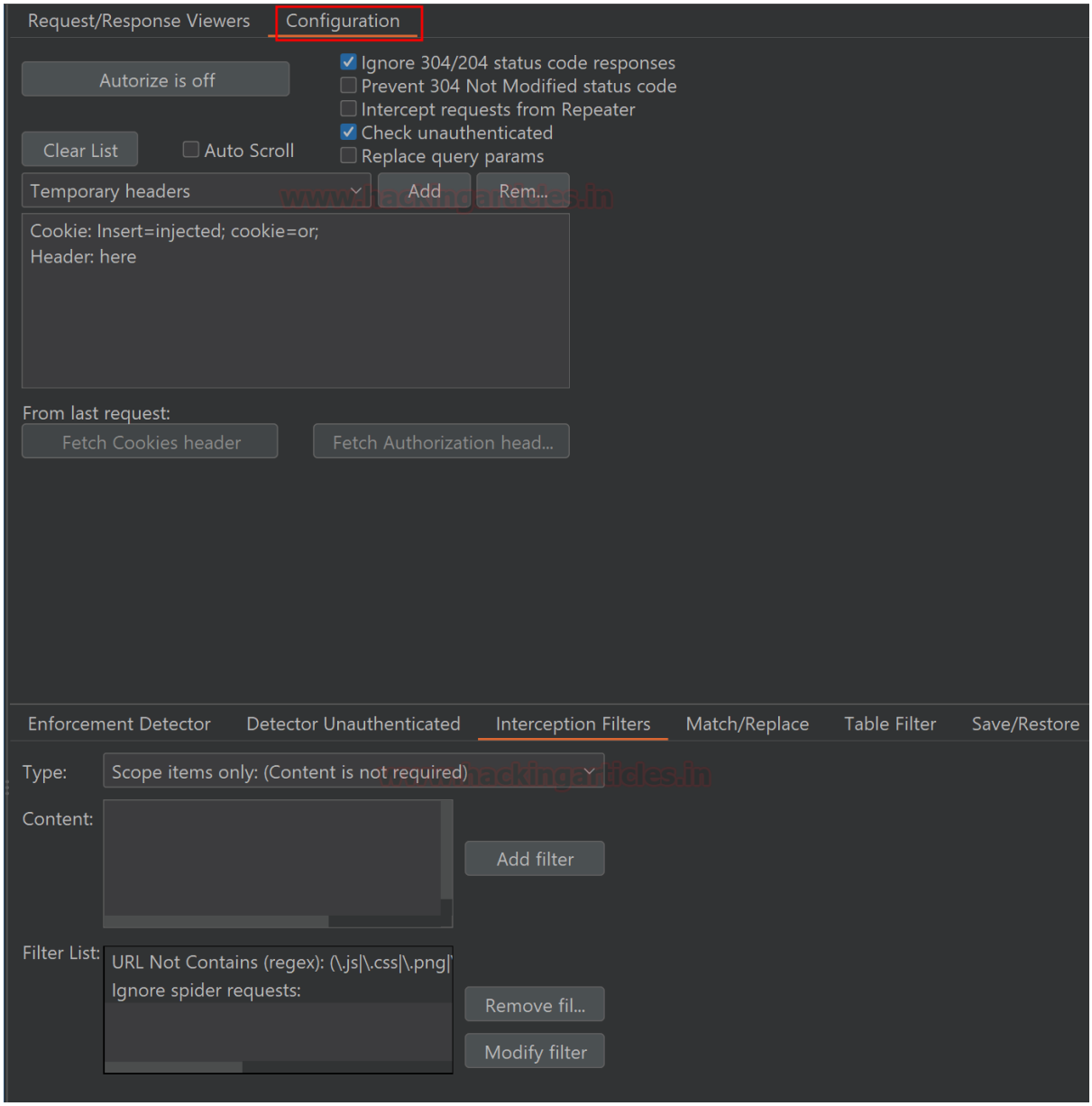


Configuration: Under the configuration tab you will see Autorize is off by default, when you are ready to capture the request first put Autorize on. There are also some configurations for capturing a request and server status code. Depending on your preference, you can select it.

Here, under the Temporary header box; you need to put the normal user token/cookies/header value that you want to replace within the actual request i.e. if any application is using a JWT token for auth mechanism you need to put that value here.

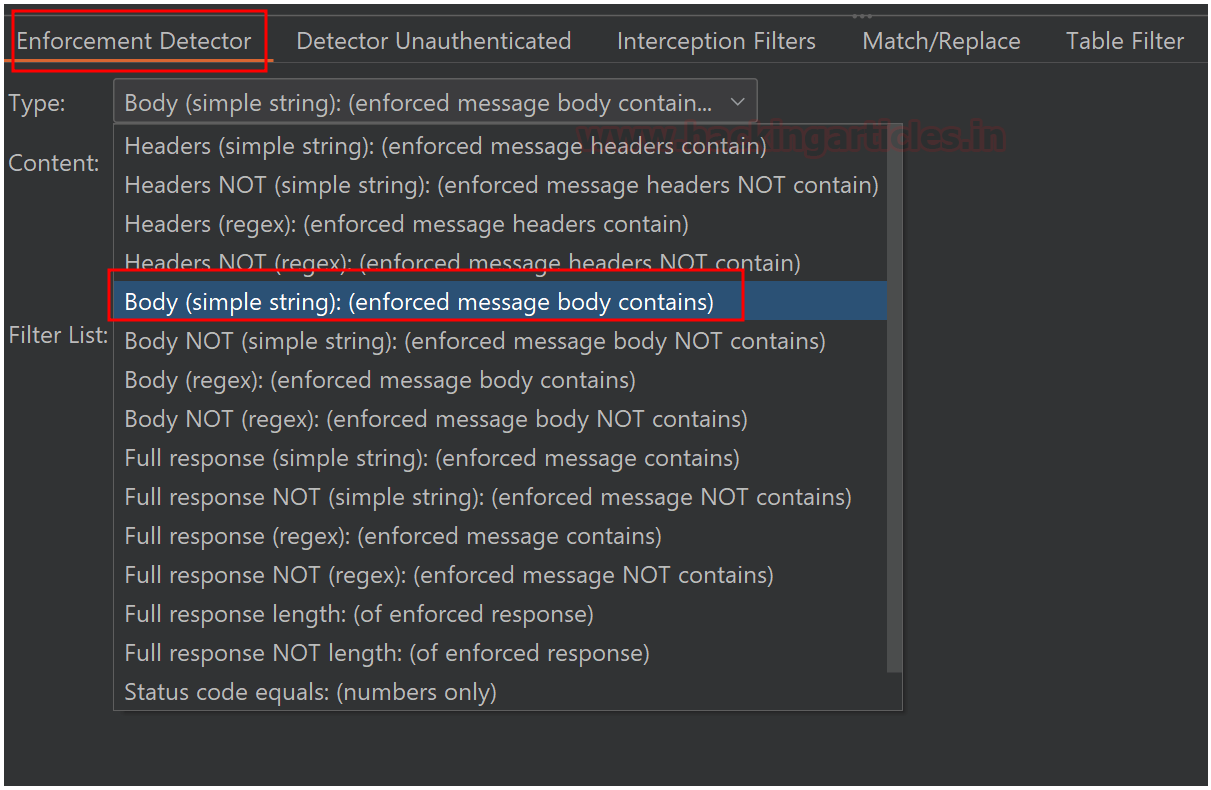
Either you can manually add the auth value or below is the option to fetch it from the last request. If you want to add the cookies header from the last request – click on 'Fetch Cookies header' or if you want to add Authorization header – click on 'Fetch Authorization header'.

Generally, the session cookies are under Cookies Header and the auth token comes under Authorization Header.



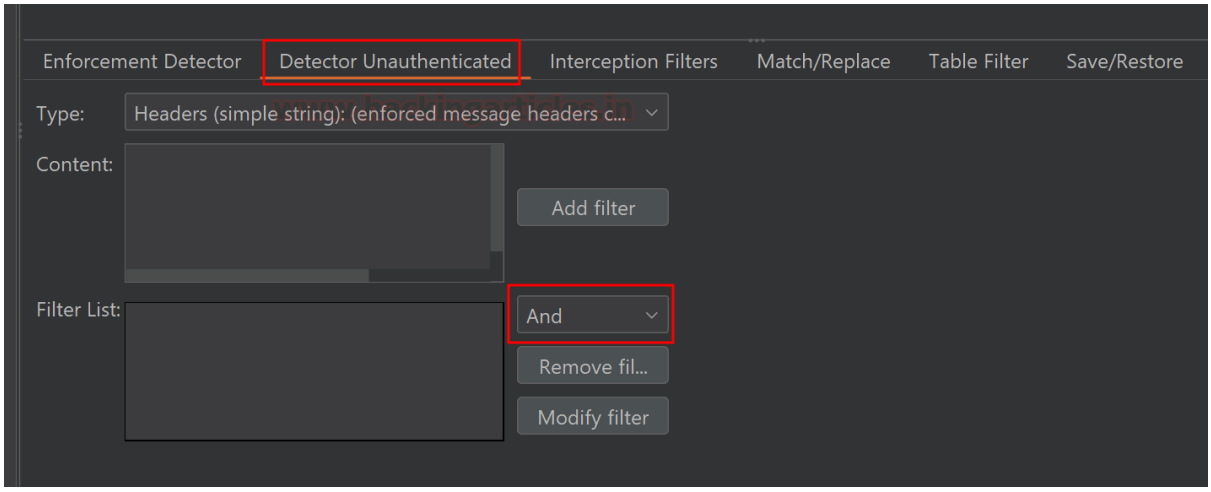
Once the session cookies are loaded, it is essential to instruct Authorize on which requests to intercept and establish the standard behaviour for the application when dealing with unauthorized requests or those with insufficient permissions.

Commencing with the Enforcement Detector, input a characteristic of the application's response that can be anticipated when a user with limited privileges tries to perform an action, they lack sufficient permissions. In my practice, I've found that utilizing the "Body (simple string): enforced message body contains" option is the simplest to set up and functions effectively. Choose the type and content that aligns with your specific needs and remember to click the "Add filter" button.

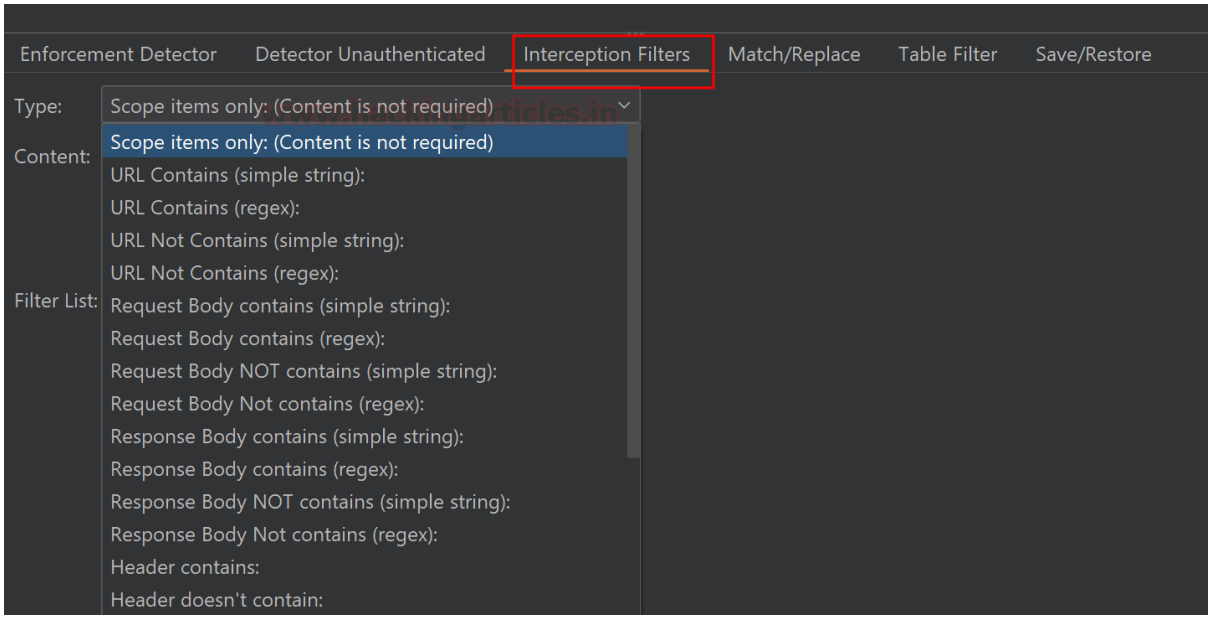


Moreover, it is necessary to understand that it automatically sets the default comparison to "And" when assessing multiple filters. Therefore, if the application generates distinct error messages, such as one for trying to read a file and another for attempting to access administrative features, you should create a filter for each scenario and switch the "And" to "Or."

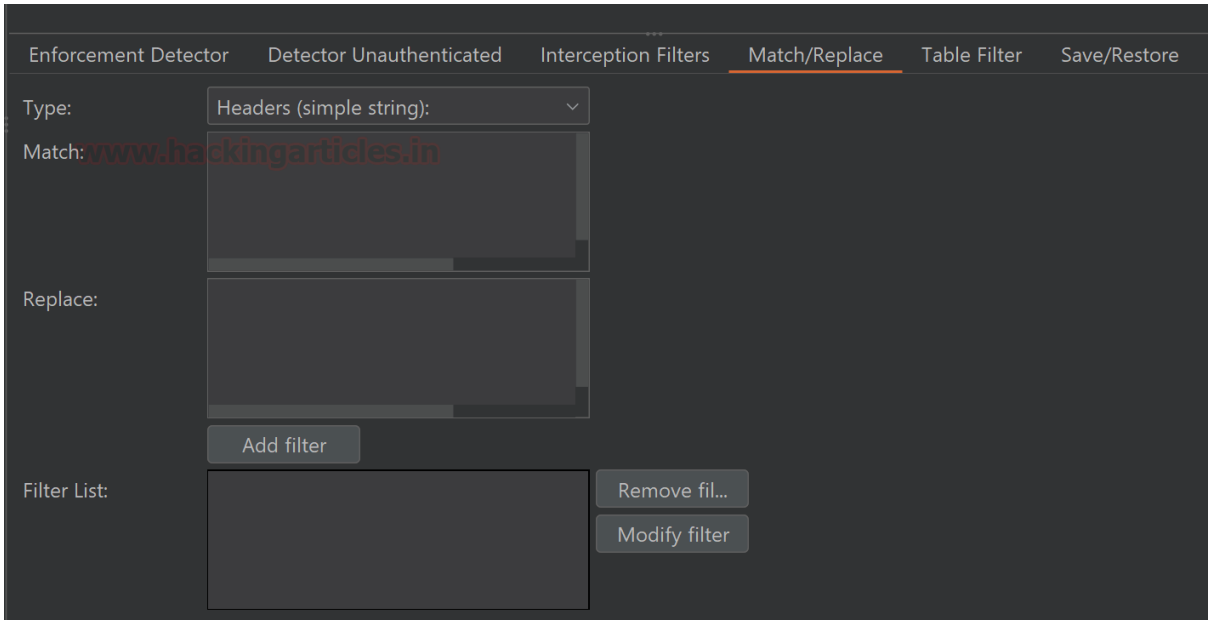
Follow the same procedure for the Unauthenticated Detector



The interception filter will intercept "Scope items only" regardless of content and from those requests, it will ignore spider requests and URLs containing image extensions. You may select on your preference and click "Add filter" when type is selected.

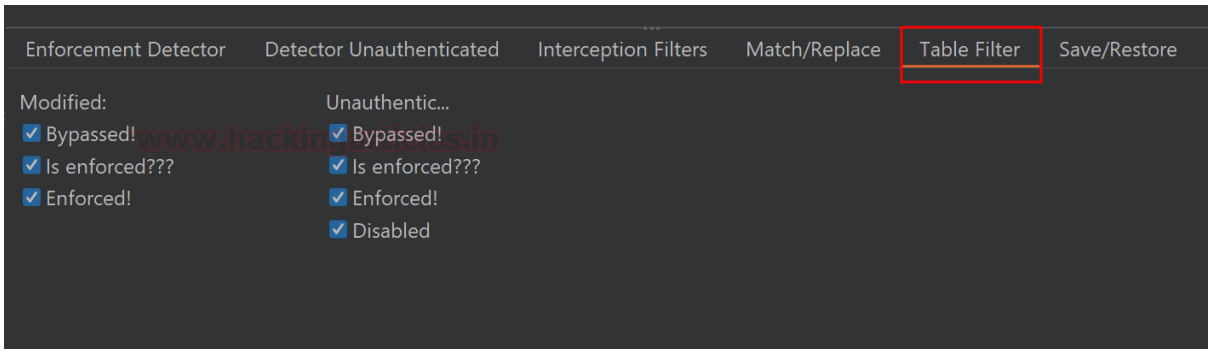


This is another additional feature Match/Replace. You can select it from this site if you need to change any specific header or body parameter on the Autorize request. Suppose there is a parameter name 'u.name' on the request body, and it has to be replaced by an Admin EID i.e.:="a.name") for proper access circumvention. You can tell Autorize via adding here.



You can select the type of requests that you want to see under the Table Filter bar,

- bypassed!: the endpoint may be vulnerable to IDOR,
- Is enforced!: endpoint seems to be protected but re-check once,
- Enforcing!: against IDOR, the endpoint is clearly protected.

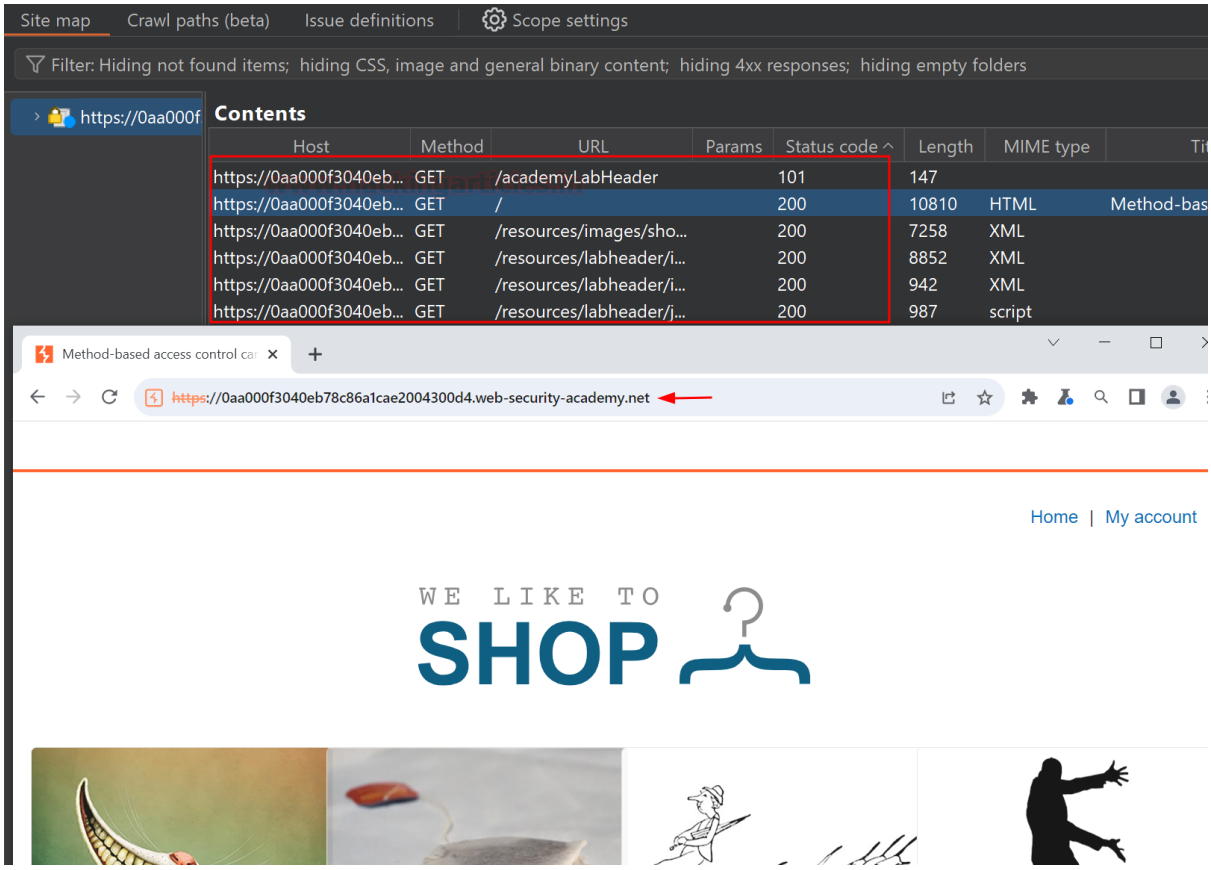


You can save and export the data for further analysis under the Save/Restore tab.

Practical Demonstration of Autorize in Action

Let's do a quick demonstration to understand in an easy way, to perform this practical we are going to use a pre-setup Port Swigger lab "Method-based access control can be circumvented". Click on access the lab and browser the application.

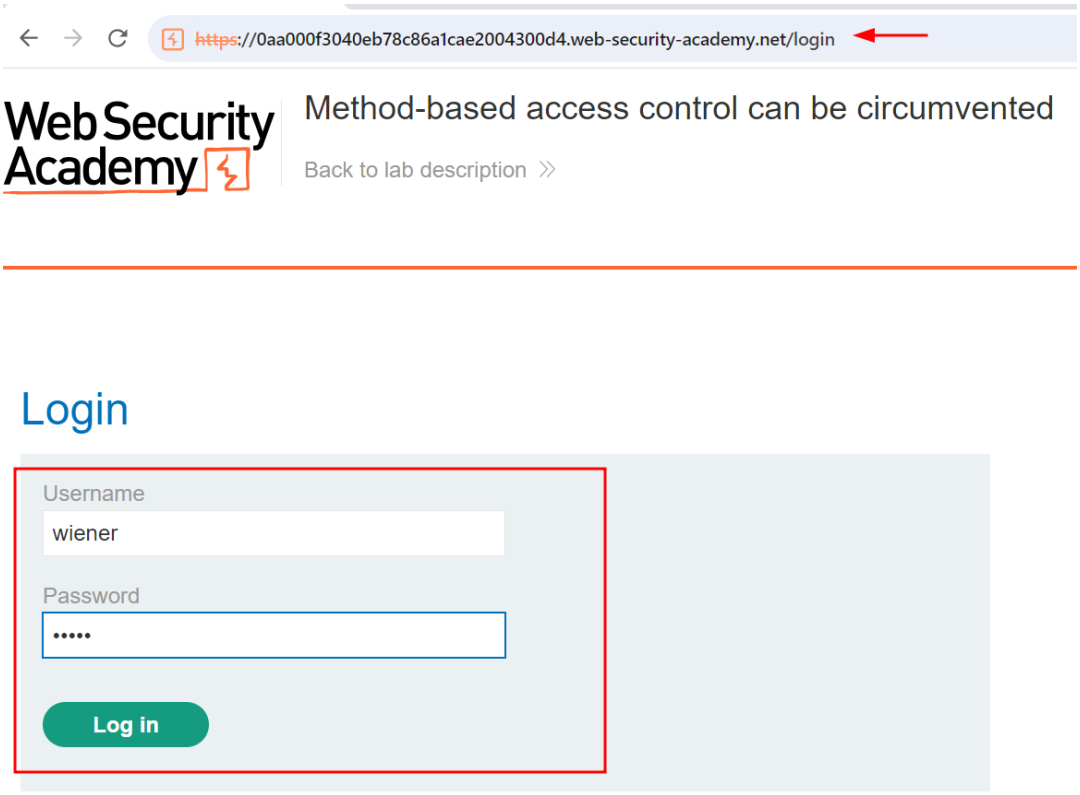
This will show a Broken Access Control vulnerability with two users that have different role higher and lower privilege users. The same concept can be applied to same-level users.



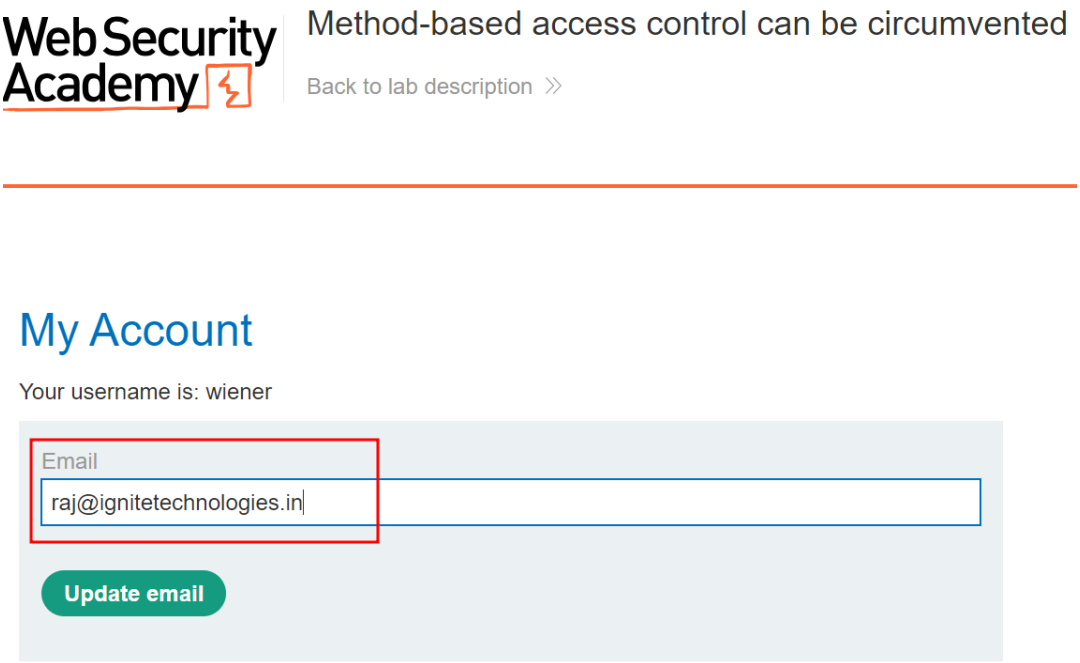
First, we have to capture the cookies for low privileged user (normal user). We are using the default normal user credentials,

Wiener:peter

And logged into the application to capture session cookie.



Updated some more details.

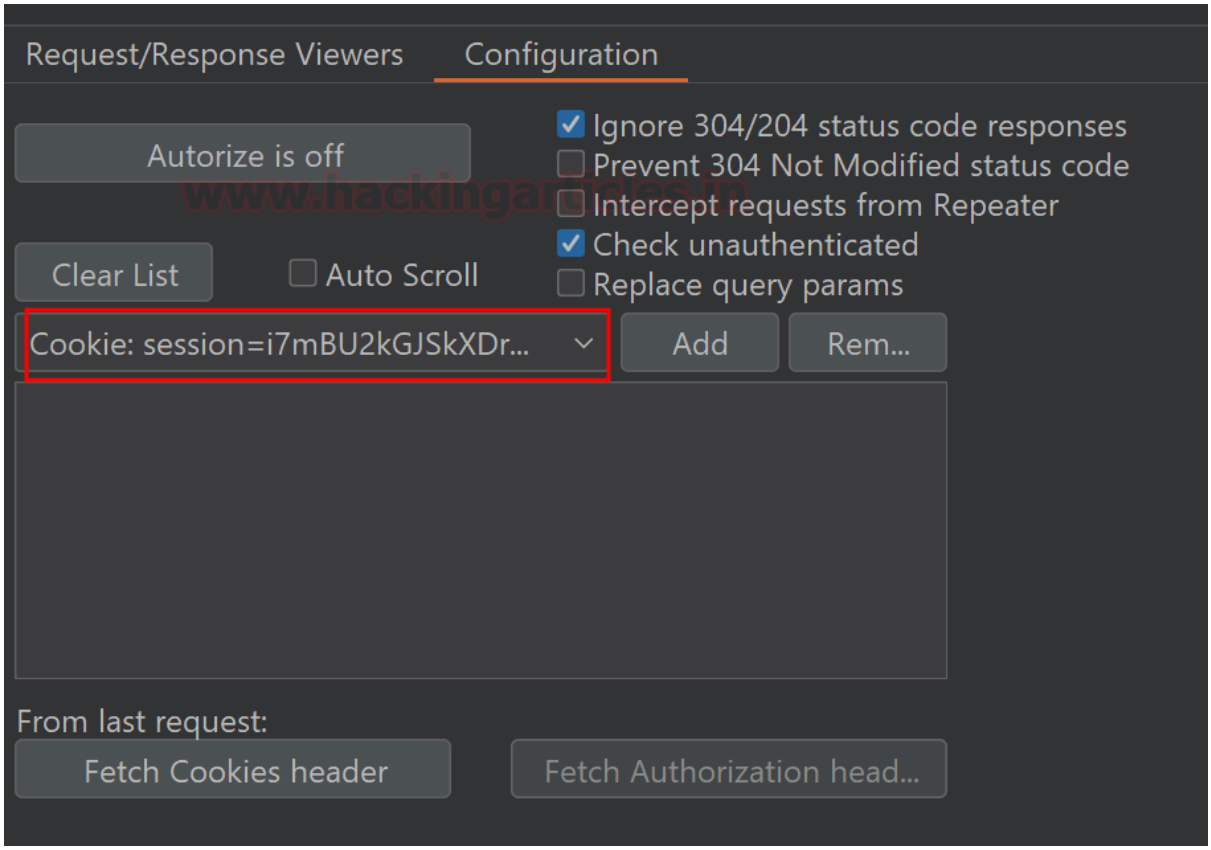


You will see the below capture session cookie in to the login request. Now copy this cookie header.

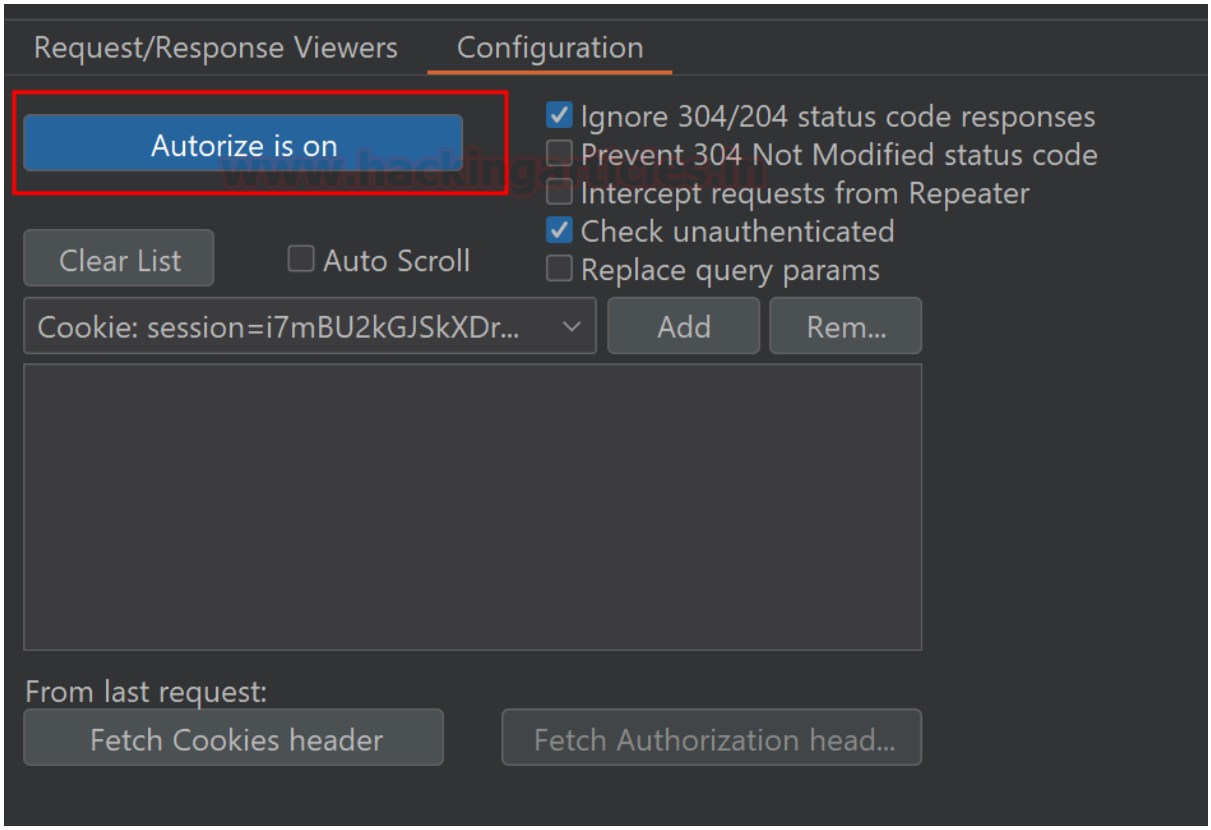
```
1 ×  +
Send  Cancel  <  >

Request
Pretty  Raw  Hex
1  POST /login HTTP/2
2  Host: 0aa000f3040eb78c86a1cae2004300d4.web-security-academy.net
3  Cookie: session=i7mBU2kGJSkXDrFhmTT1XuKokr63zvKx
4  Content-Length: 30
5  Cache-Control: max-age=0
6  Sec-Ch-Ua:
7  Sec-Ch-Ua-Mobile: ?0
8  Sec-Ch-Ua-Platform: ""
9  Upgrade-Insecure-Requests: 1
10 Origin: https://0aa000f3040eb78c86a1cae2004300d4.web-security-academy.net
11 Content-Type: application/x-www-form-urlencoded
12 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko
13 Accept:
   text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;
   0.7
14 Sec-Fetch-Site: same-origin
15 Sec-Fetch-Mode: navigate
16 Sec-Fetch-User: ?1
17 Sec-Fetch-Dest: document
18 Referer: https://0aa000f3040eb78c86a1cae2004300d4.web-security-academy.net/login
19 Accept-Encoding: gzip, deflate, br
20 Accept-Language: en-US,en;q=0.9
21
22 username=wiener&password=peter
```

Add this cookie header value to Autorize tab as shown below,

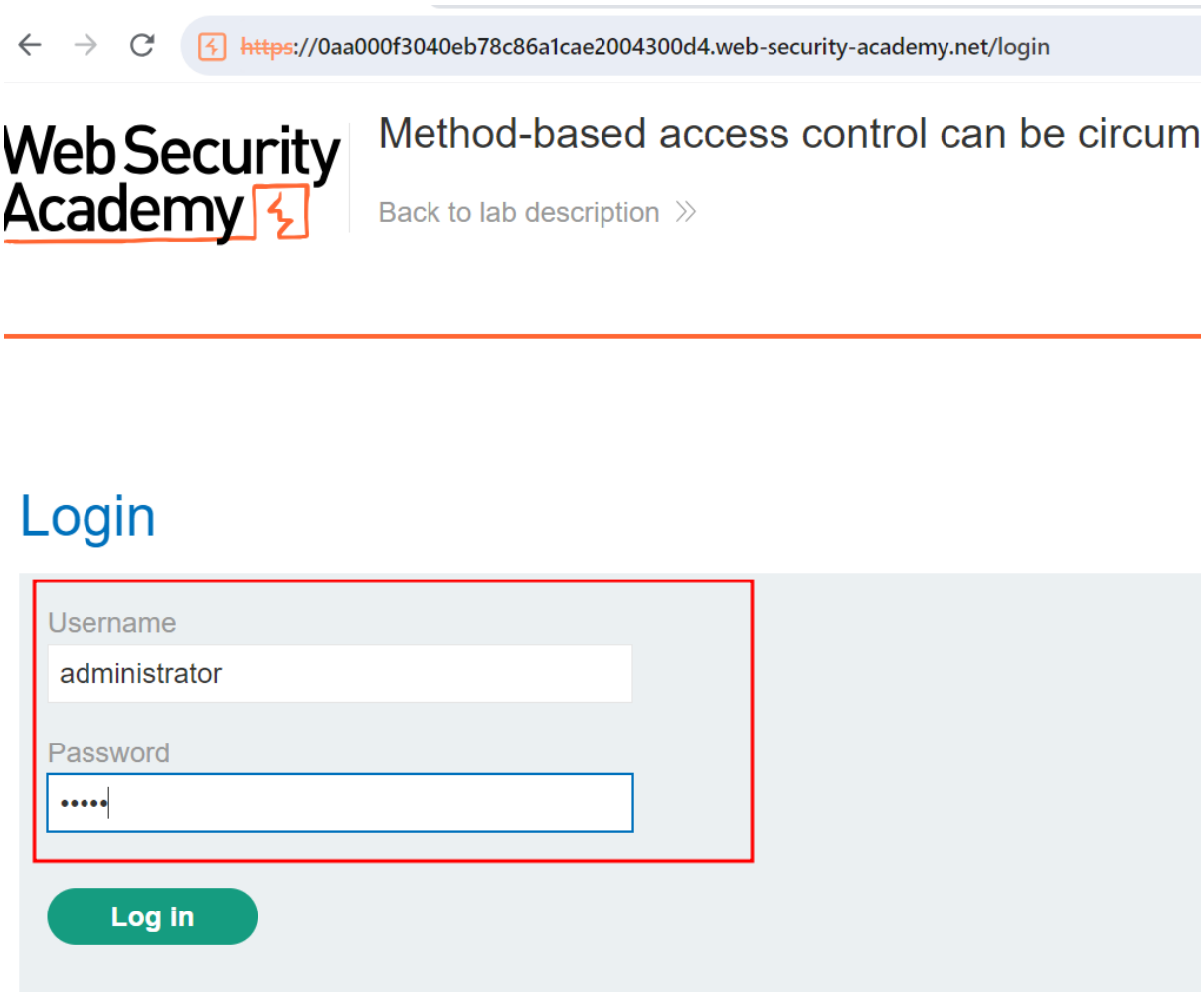


And keep Autorize on.



In order to, check the auth bypass now we have to log in with high privilege (admin user). Go to login page again and use admin credentials to log in,

Administrator:admin



After successfully logging in and browsing the all admin-only URLs. You can see under the Autorize tab some highlighted requests

The **Authz. Status** indicates which endpoints are accessible to wiener (normal user).

The **Unauth. Status** pertains to unauthorized users, effectively eliminating the cookie and all authorization headers. You can opt to disable this feature by deselecting the "Check unauthenticated" option in the Autorize configuration tab.

Red [Bypassed!] : endpoint could be vulnerable to access control/IDOR issues.

Orange [Is enforced!] : endpoint seems to be protected but cross-check manually by replacing the cookies value.

Green [Enforced!] : endpoint is clearly protected against access control/IDOR issues.

...	...	URL	...	...	...	Authz. Status	Unauth. Status
1	...	https://0aa000f3040eb78c86a1cae2004300d4.web-security-academy.net:443/login	0	0	0	Bypassed!	Bypassed!
2	...	https://0aa000f3040eb78c86a1cae2004300d4.web-security-academy.net:443/my-account?id=administrator	...	...	...	Bypassed!	Bypassed!
3	...	https://0aa000f3040eb78c86a1cae2004300d4.web-security-academy.net:443/academyLabHeader	0	...	...	Enforced!	Enforced!
4	...	https://0aa000f3040eb78c86a1cae2004300d4.web-security-academy.net:443/admin	...	...	...	Bypassed!	Bypassed!
5	...	https://0aa000f3040eb78c86a1cae2004300d4.web-security-academy.net:443/academyLabHeader	0	...	...	Enforced!	Enforced!
6	...	https://0aa000f3040eb78c86a1cae2004300d4.web-security-academy.net:443/admin-roles	0	0	0	Bypassed!	Bypassed!
7	...	https://0aa000f3040eb78c86a1cae2004300d4.web-security-academy.net:443/admin	...	...	...	Bypassed!	Bypassed!
8	...	https://0aa000f3040eb78c86a1cae2004300d4.web-security-academy.net:443/academyLabHeader	0	...	...	Enforced!	Enforced!
9	...	https://0aa000f3040eb78c86a1cae2004300d4.web-security-academy.net:443/admin-roles	0	0	0	Bypassed!	Bypassed!
...	...	https://0aa000f3040eb78c86a1cae2004300d4.web-security-academy.net:443/admin	...	...	...	Bypassed!	Bypassed!
...	...	https://0aa000f3040eb78c86a1cae2004300d4.web-security-academy.net:443/academyLabHeader	0	...	...	Enforced!	Enforced!

As visible in above image, request 1, 2, 6, and 7 are having Broken access control issue.

Keep in mind that do not blindly follow the Autorize result, The Red highlight requests do not mean that all endpoints are vulnerable or bypassed. There may be false positives; You must do a cross-check.

Some other possible scenarios, suppose you are testing auth issues with the two same level of users. As a result, you will see Authz. Status shows Bypassed! And Unauth. Status shows Enforced! In that case improper authorization can be found on the request which shows that the specific endpoint can be accessed by the 2nd user but has correctly implemented authorization for any unauthorized users.

When you select any highlighted request, on the right side you will see the detailed information about modified, original & unauthenticated request and responses.

Organizer Extensions Learn Autorize

Request/Response Viewers Configuration

Modified Request Modified Response Expand

Pretty Raw ex

```
1 POST /my-account/change-email HTTP/2
2 Host: 0a2000f4041f53818018353200cb00fd.web-security-academy.net
3 Content-Length: 34
4 Cache-Control: max-age=0
5 Sec-Ch-Ua:
6 Sec-Ch-Ua-Mobile: ?0
7 Sec-Ch-Ua-Platform: ""
8 Upgrade-Insecure-Requests: 1
9 Origin: https://0a2000f4041f53818018353200cb00fd.web-security-academy.net
10 Content-Type: application/x-www-form-urlencoded
11 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML,
12 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/w
13 Sec-Fetch-Site: same-origin
14 Sec-Fetch-Mode: navigate
15 Sec-Fetch-User: ?1
16 Sec-Fetch-Dest: document
17 Referer: https://0a2000f4041f53818018353200cb00fd.web-security-academy.net/my-ac
18 Accept-Encoding: gzip, deflate, br
19 Accept-Language: en-US,en;q=0.9
20 Cookie: session=eO2BUCQvmsEIqb87XJBCiACSHSdXD9Y5
```

Search

Original Request Original Response Expand

Pretty Raw Hex

```
1 POST /my-account/change-email HTTP/2
2 Host: 0a2000f4041f53818018353200cb00fd.web-security-academy.net
3 Cookie: session=dxqZCSrDns2hwlOFoRl866ROgo4Gv6Cb
4 Content-Length: 34
5 Cache-Control: max-age=0
6 Sec-Ch-Ua:
7 Sec-Ch-Ua-Mobile: ?0
8 Sec-Ch-Ua-Platform: ""
9 Upgrade-Insecure-Requests: 1
10 Origin: https://0a2000f4041f53818018353200cb00fd.web-security-academy.net
11 Content-Type: application/x-www-form-urlencoded
12 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML,
13 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/w
14 Sec-Fetch-Site: same-origin
15 Sec-Fetch-Mode: navigate
16 Sec-Fetch-User: ?1
17 Sec-Fetch-Dest: document
18 Referer: https://0a2000f4041f53818018353200cb00fd.web-security-academy.net/my-ac
19 Accept-Encoding: gzip, deflate, br
20 Accept-Language: en-US,en;q=0.9
```

Search

Unauthenticated Request Unauthenticated Response Expand

Pretty Raw Hex

```
1 POST /my-account/change-email HTTP/2
2 Host: 0a2000f4041f53818018353200cb00fd.web-security-academy.net
3 Content-Length: 34
4 Cache-Control: max-age=0
5 Sec-Ch-Ua:
6 Sec-Ch-Ua-Mobile: ?0
7 Sec-Ch-Ua-Platform: ""
8 Upgrade-Insecure-Requests: 1
9 Origin: https://0a2000f4041f53818018353200cb00fd.web-security-academy.net
10 Content-Type: application/x-www-form-urlencoded
11 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML,
12 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/w
13 Sec-Fetch-Site: same-origin
14 Sec-Fetch-Mode: navigate
15 Sec-Fetch-User: ?1
16 Sec-Fetch-Dest: document
17 Referer: https://0a2000f4041f53818018353200cb00fd.web-security-academy.net/my-ac
18 Accept-Encoding: gzip, deflate, br
19 Accept-Language: en-US,en;q=0.9
20
21 email=rraj%40ignitetechnologies.in
```

Search

Burpsuite for Pentester: Logger++

Suppose you are a web explorer, and you want to know everything about a website. Burp Logger++ is like your trusty notebook. It is super helpful because it has a magical filter. You can tell it what kind of information you are looking for, and it will only show you those things.

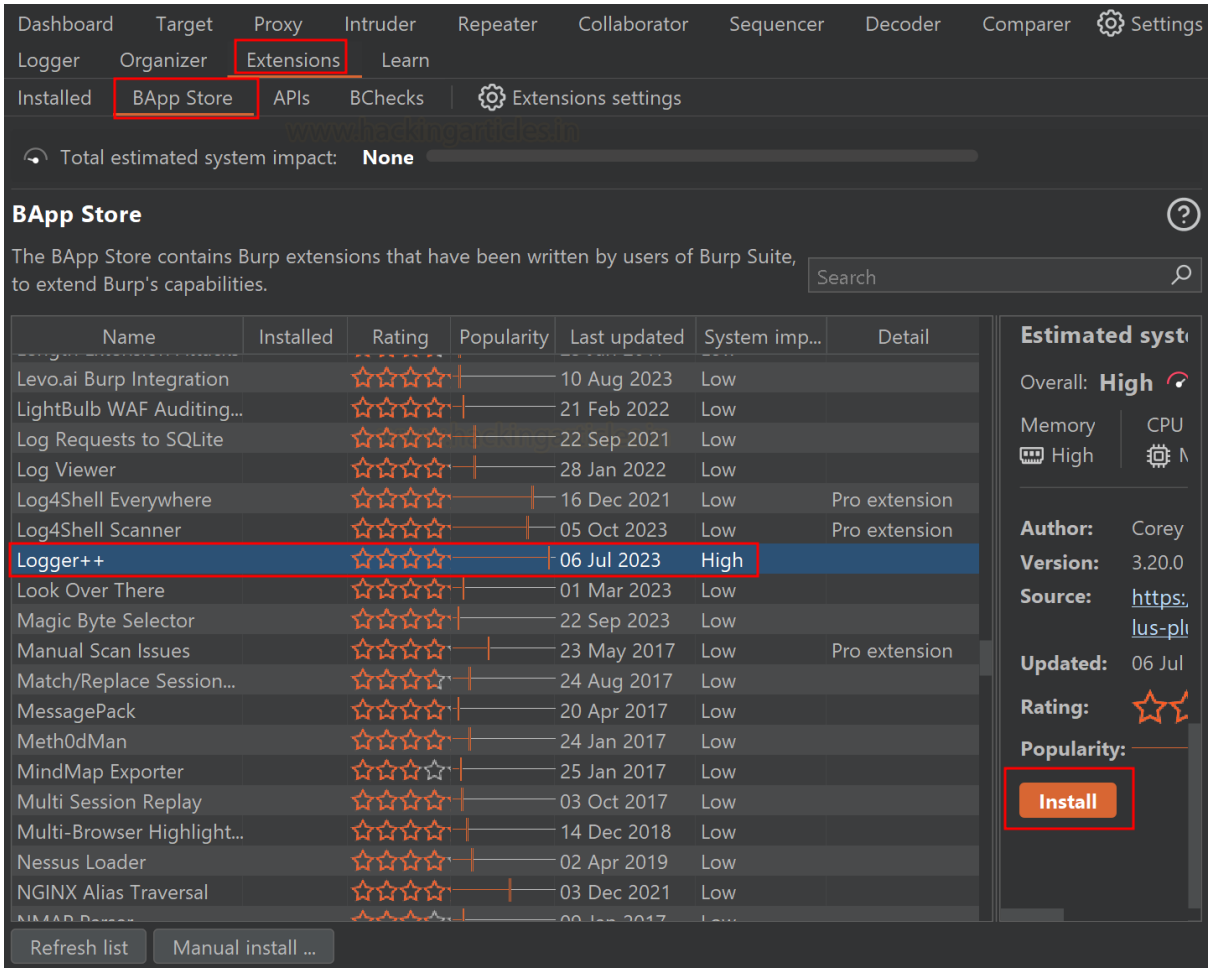
With Burp Logger++, you can also color-code things. Think of it like using different colors to highlight the most important parts of a picture. This helps you spot the important stuff quickly.

- **Setting Up & Navigating**
- **Query-Based Filter**
- **Filter Library**
- **Regex-Based Filter**
- **Export Data Feature**

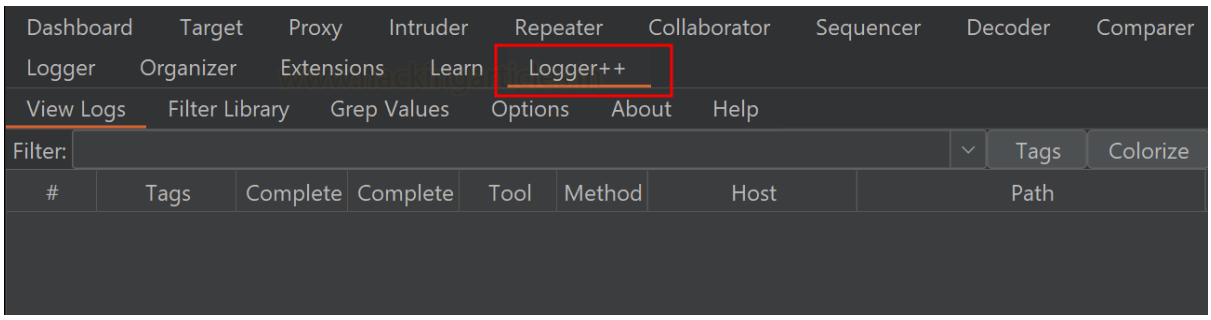
Setting Up & Navigating

You can download and install the extension from the BApp Store. Go to Extensions > Bapp Store. Here, search for Logger++ or simply scroll down.

Click on it, on the right side scroll down and install it.

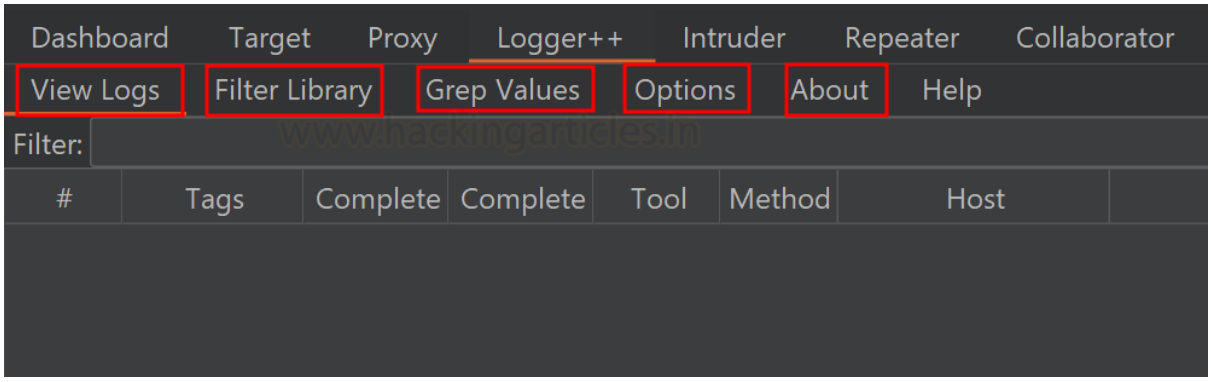


After successful installation, it will appear on the toolbar.

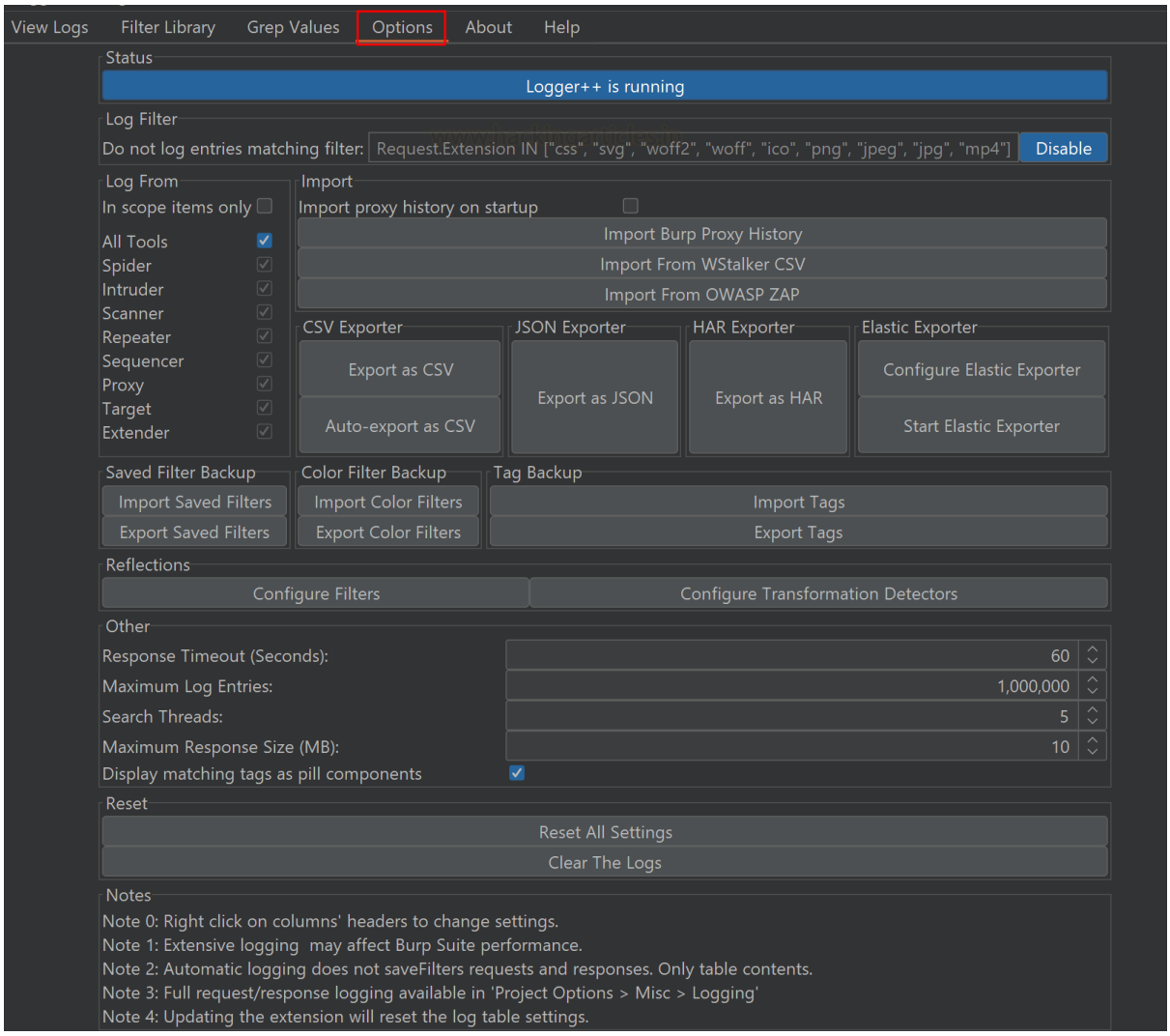


Navigating

There are a lot of options visible to you. First, let's explore the "Options" tab to discover what advanced settings are included in this extension.



Navigate to "Options" to see the various log filter options. It allows you to customize logging setting as per your preference.



Logger++ is running by default. Here are some other important settings:

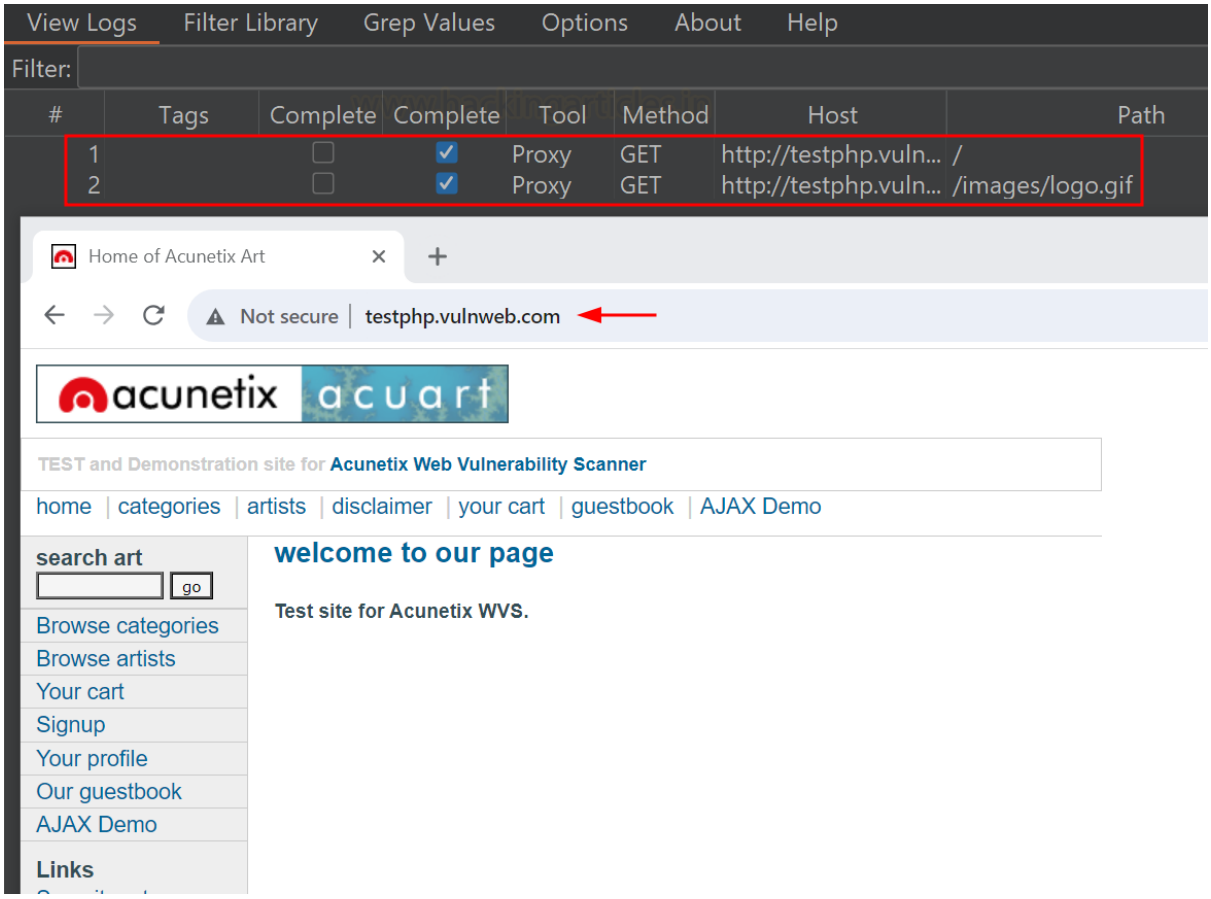
- Log Filter: This feature lets you specifically choose the requests that you don't need to record for analysis, or you may turn it off when not in use.
- Log From: It enables you to capture data from the specific logs that you want to capture.

- Import: You can import log data from CSV and OWASP ZAP reports with this function.
- Export: The log data can be exported for further analysis.

Depending on your preferences, you can use different configurations. We are sticking with the default settings for the time being.

Query-Based Filter

The View Log tab contains all the logs. Using this website "vulnweb" as an example, browse it and simply scan the entire site; all logs will show up here under the View Logs page.

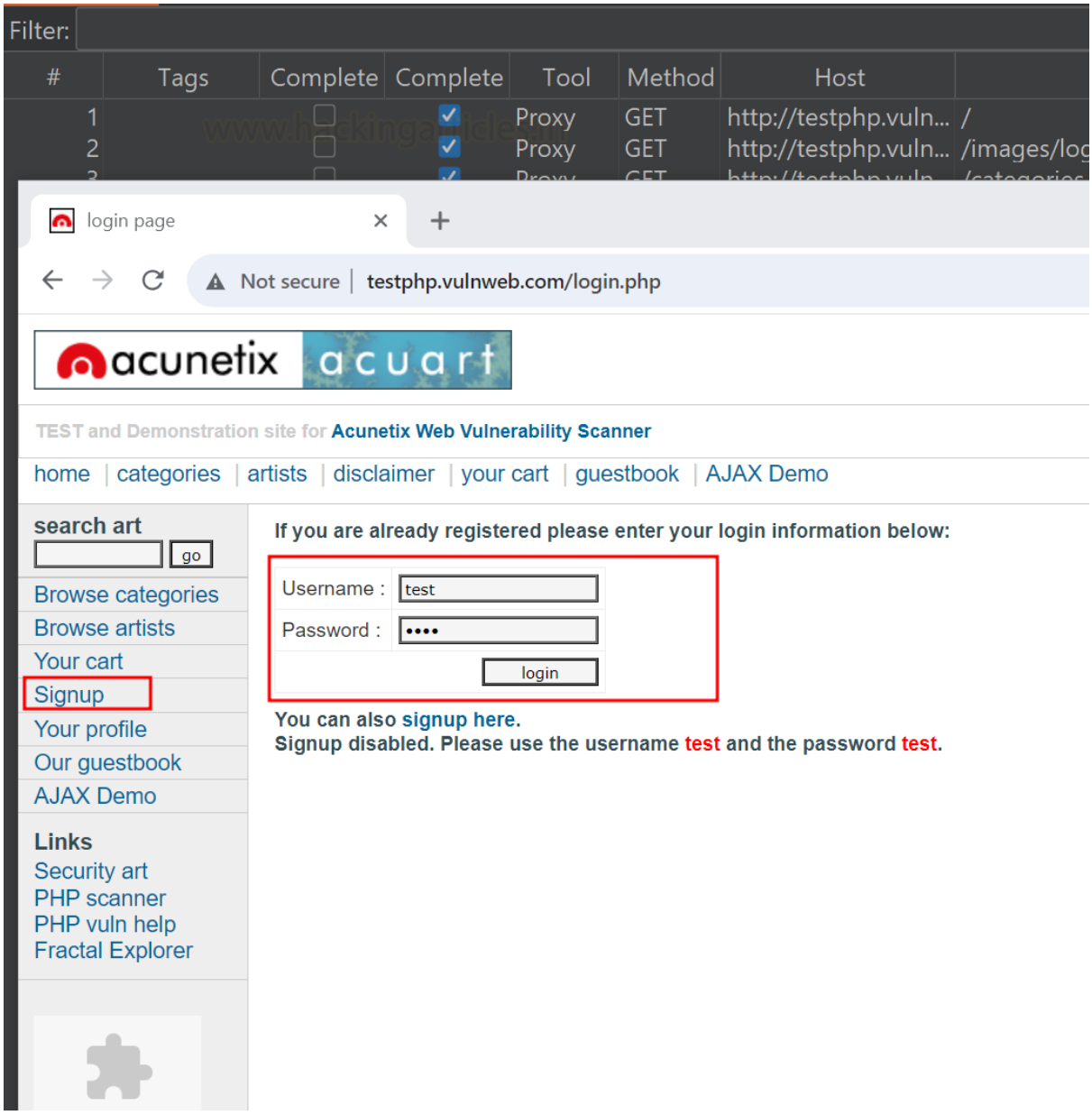


Now, go to Signup. To capture the logs for credentials, enter the test login details.

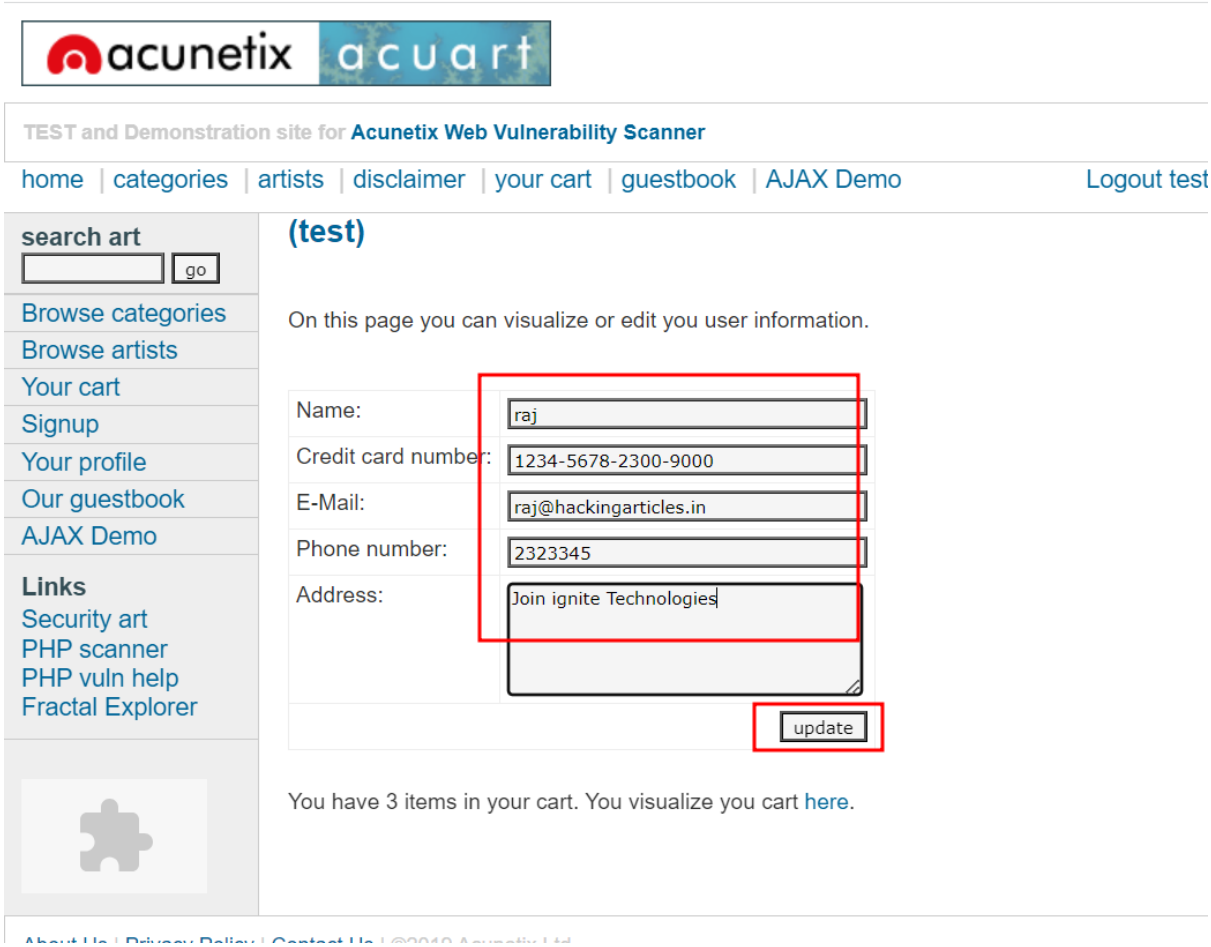
Username: test

Password: test

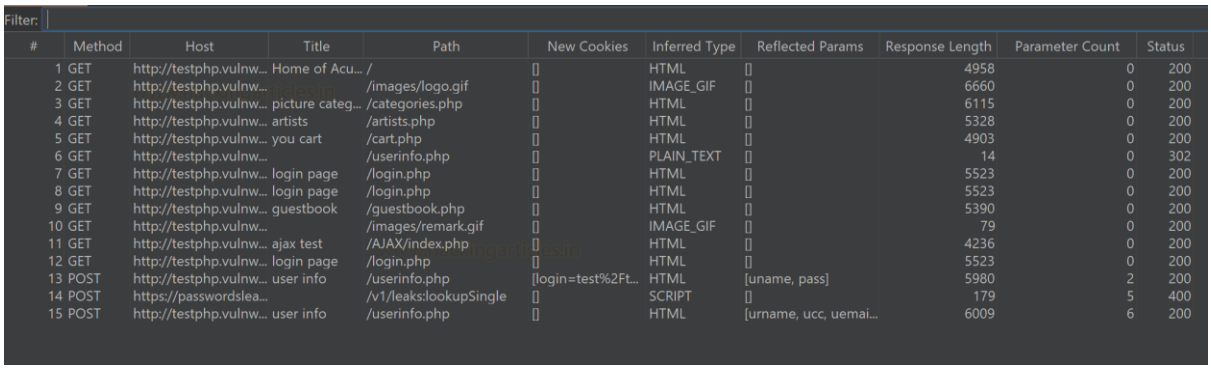
then click on "Login".



Let's update some more details to capture more requests for further analysis.



You can see that all requests have been captured here in View Logs.



Filter:

#	Method	Host	Title	Path	New Cookies	Inferred Type	Reflected Params	Response Length	Parameter Count	Status
1	GET	http://testphp.vulnw...	Home of Acu...	/	[]	HTML	[]	4958	0	200
2	GET	http://testphp.vulnw...		/images/logo.gif	[]	IMAGE_GIF	[]	6660	0	200
3	GET	http://testphp.vulnw...	picture categ...	/categories.php	[]	HTML	[]	6115	0	200
4	GET	http://testphp.vulnw...	artists	/artists.php	[]	HTML	[]	5328	0	200
5	GET	http://testphp.vulnw...	you cart	/cart.php	[]	HTML	[]	4903	0	200
6	GET	http://testphp.vulnw...		/userinfo.php	[]	PLAIN_TEXT	[]	14	0	302
7	GET	http://testphp.vulnw...	login page	/login.php	[]	HTML	[]	5523	0	200
8	GET	http://testphp.vulnw...	login page	/login.php	[]	HTML	[]	5523	0	200
9	GET	http://testphp.vulnw...	guestbook	/guestbook.php	[]	HTML	[]	5390	0	200
10	GET	http://testphp.vulnw...		/images/remark.gif	[]	IMAGE_GIF	[]	79	0	200
11	GET	http://testphp.vulnw...	ajax test	/AJAX/index.php	[]	HTML	[]	4236	0	200
12	GET	http://testphp.vulnw...	login page	/login.php	[]	HTML	[]	5523	0	200
13	POST	http://testphp.vulnw...	user info	/userinfo.php	[login=test%2Ft...	HTML	[uname, pass]	5980	2	200
14	POST	https://passwordslea...		/v1/leaks:lookupSingle	[]	SCRIPT	[]	179	5	400
15	POST	http://testphp.vulnw...	user info	/userinfo.php	[]	HTML	[urname, ucc, uemai...	6009	6	200

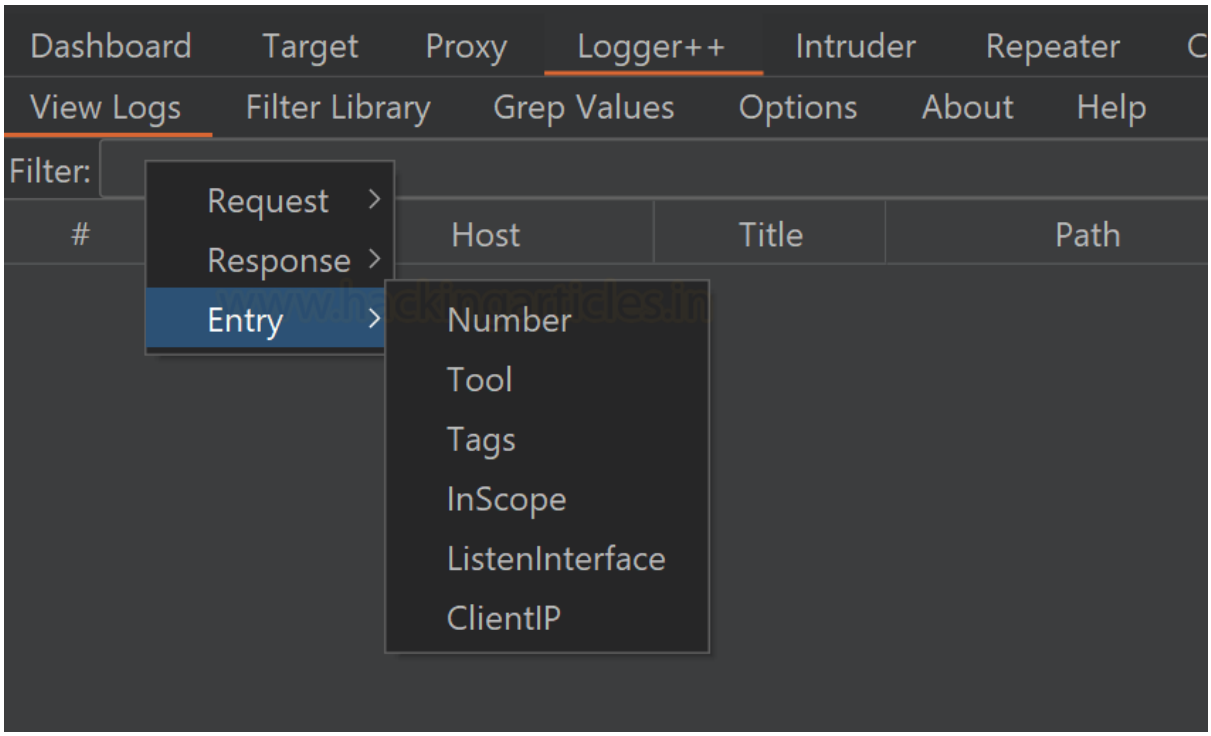
Magical Filter

You can use filter to selectively view or manipulate HTTP requests and responses. These filters help you focus on specific aspects of the web traffic and are especially useful during

security testing. The working is based on query string. It accepts a logical query and returns output based on them.

You have some advanced choices with the filter options:

Entry: You can apply filters according to number, tool, tags, InScope, and other criteria.



Request: It lets you filter just the request itself using many options such as header, body, URL, method, parameters, cookies, etc. As shown below:

View Logs | Filter Library | Grep Values | Options | About | Help

Filter:

#	Method	Host	
1	GET	http://testphp.v	
2	GET	http://testphp.v	
3	GET	http://testphp.vulnw...	picture c
4	GET	http://testphp.vulnw...	artists
5	GET	http://testphp.vulnw...	you cart
6	GET	http://testphp.vulnw...	
7	GET	http://testphp.vulnw...	login pag
8	GET	http://testphp.vulnw...	login pag
9	GET	http://testphp.vulnw...	guestboo
10	GET	http://testphp.vulnw...	
11	GET	http://testphp.vulnw...	ajax test
12	GET	http://testphp.vulnw...	login pag
13	POST	http://testphp.vulnw...	user info
14	POST	https://passwordslea...	
15	POST	http://testphp.vulnw...	user info

Entry
Request
Response

AsBase64
Headers
Body
BodyLength
Time
Length
Tool
Comment
Complete
URL
Method
Path
Query
PathQuery
Protocol
IsSSL
UsesCookieJar
Hostname
Host
Port
ContentType
RequestHttpVersion
Extension
Referrer
HasParams
HasGetParam
HasPostParam
HasSentCookies
CookieString
ParameterCount
Parameters
Origin

Pretty | Raw | Hex

1

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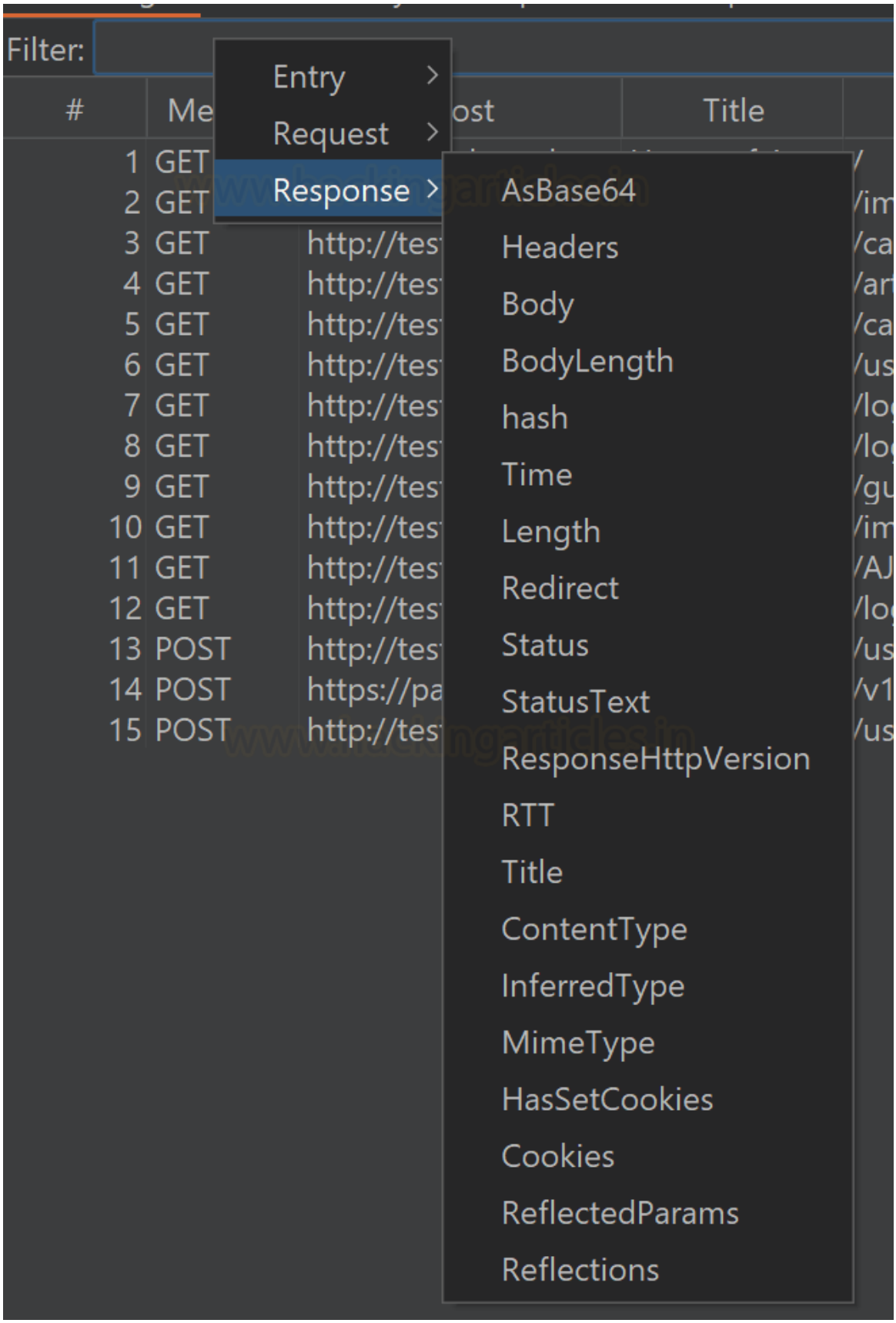
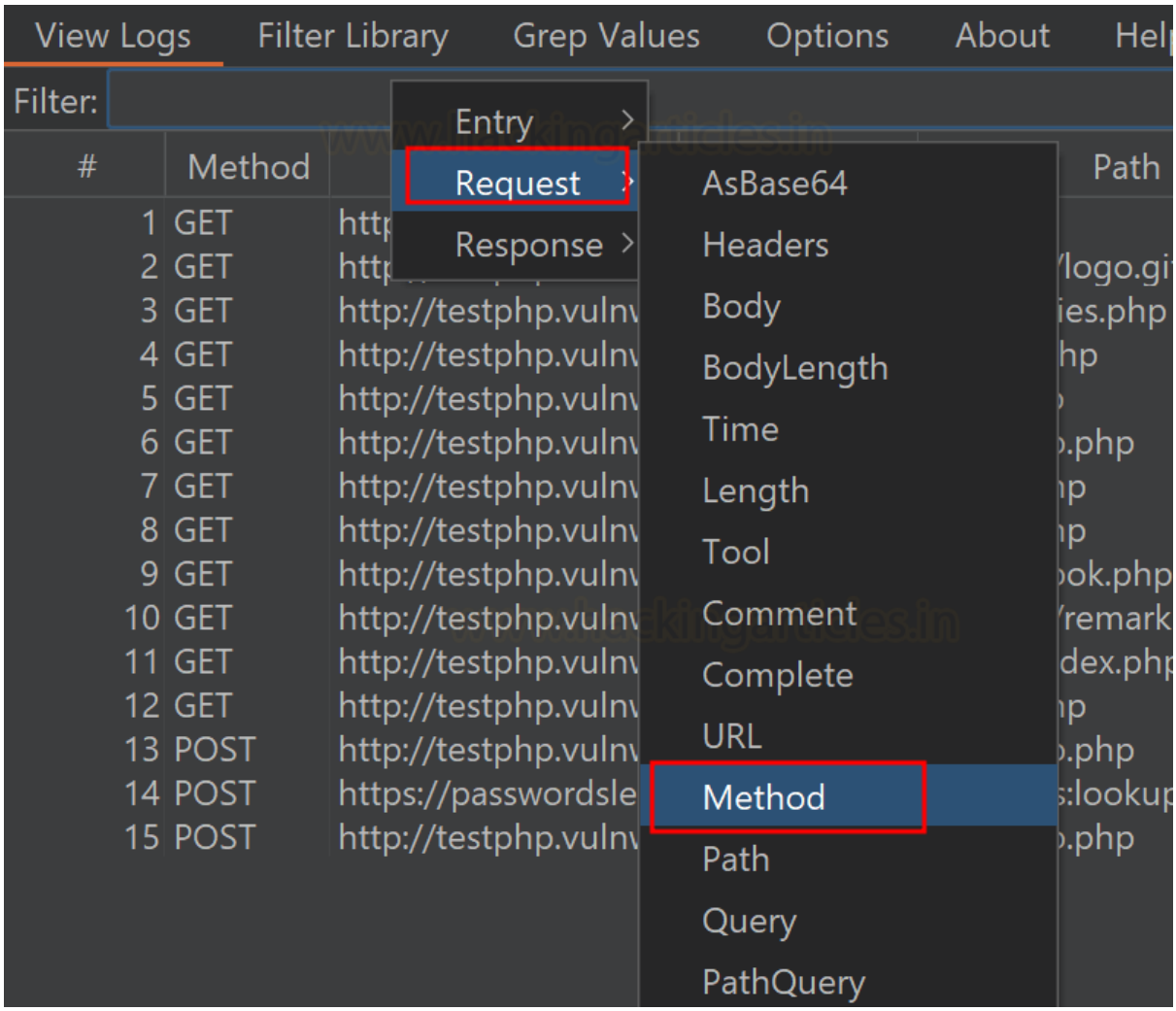
Response: It lets you filter just the response by using various options such as header, body, Inferred Type, Method, Parameters, cookies etc. As shown below:



Scenario 1: Let's suppose you just want to view HTTP POST requests from all logs. It is understood that HTTP POST parameters are in HTTP Request.

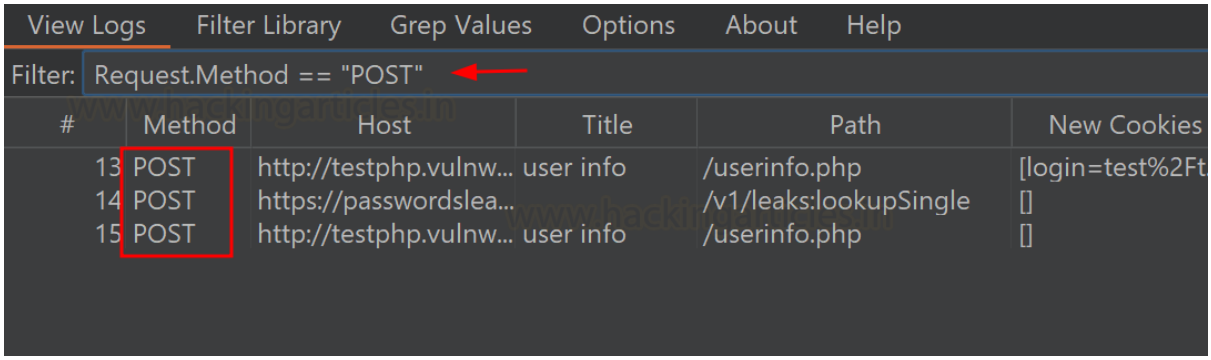
Go to Filter bar > right click > Select Request > Select Method

The method has been chosen and is visible in the filter bar.




```
Request.Method == "POST"
```

And hit enter. As result, Only HTTP POST Method requests appear.

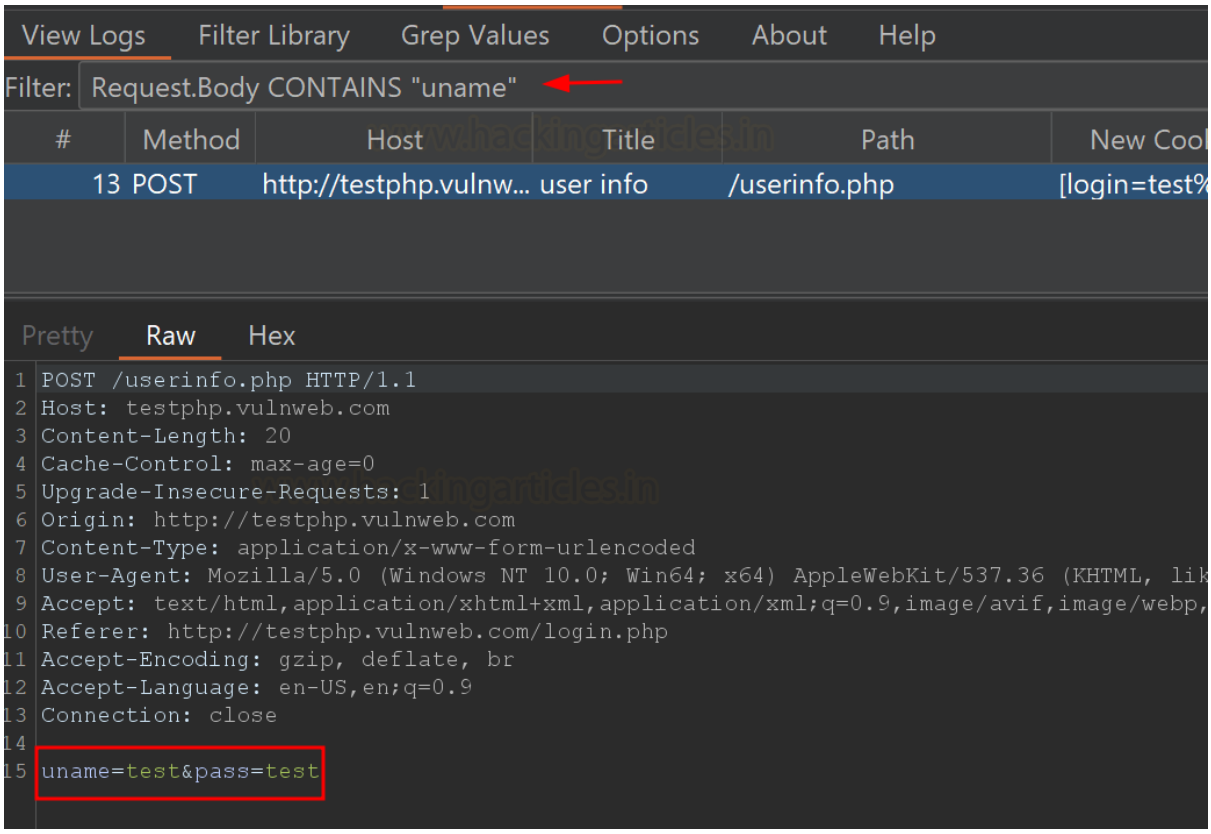


Scenario 2: Taking another example, suppose we just want to view the requests which contains any username information from all logs.

Go to Filter bar > right click > Select Request > Select Body

Request.Body CONTAINS "uname"

As a result, the following request is highlighted:



Below are some useful queries which are helpful during penetration testing.

JSON Injection (Check for only JSON request)

Response.InferredType == "json"

Injections Attack (Check for HTML, XML, JSON)

Response.InferredType IN ["json", "html", "xml"]

Sensitive File Exposed

Response.Body CONTAINS [".git", ".config", ".zip", ".swf", ".doc", ".pdf", ".xlsx", ".csv",]

Sensitive Path Exposed

Request.Path CONTAINS ["/git", "/etc", "/var"]

Request.Path MATCHES "/account*"

Sensitive Parameter in Query String

Request.Path CONTAINS ["id", "username", "password", "role", "IsAdmin"]

Sensitive Parameter in Request

Request.Body CONTAINS ["id", "username", "password", "token", "role", "EnterpriseID", "IsAdmin"]

Server Information Disclosed

Response.header CONTAINS "Server:"

CORS Misconfiguration

Response.Header MATCHES "Access-Control-Allow-Origin: *"

Check for CSRF Token

Request.Method == "POST" AND Request.Body CONTAINS "csrf"

Missing Robots.txt

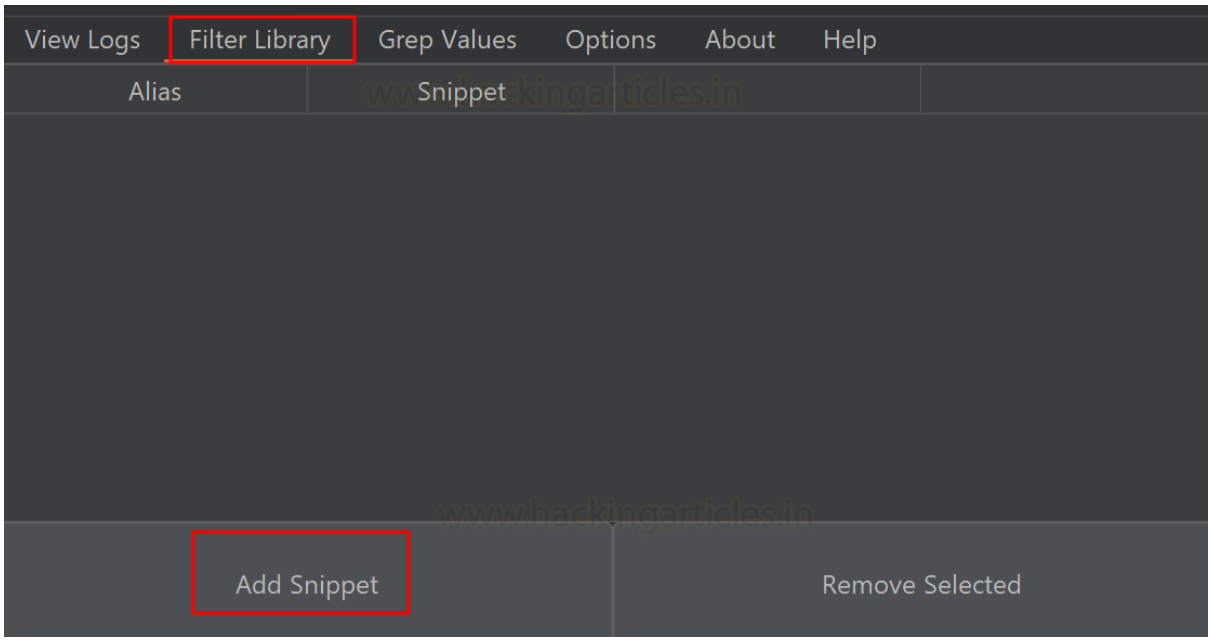
Request.Path MATCHES "/robots.txt"

URL Redirection

Request.Path CONTAINS ["redirect=", "page=", "url=", "index.page="]

Filter Library

We can use the saved or pre-configured filters from the library directly with the help of the Filter Library. When you start testing, you do not have to manually type or remember the query string of filter pattern.

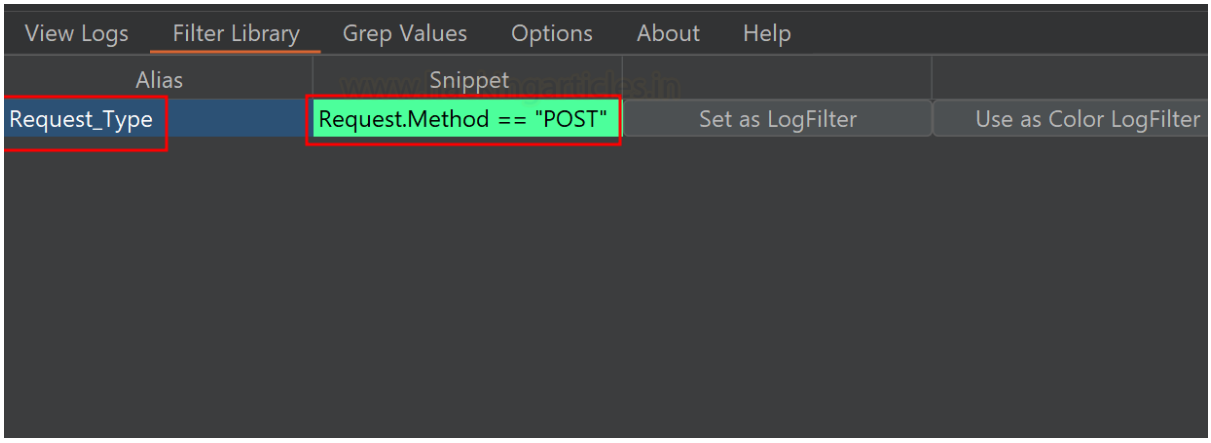


Click on "Add Snippet". Here are two values that must be added.

- **Alias**: Put any Alias name for your query string.
- **Snippet**: Add query string here.

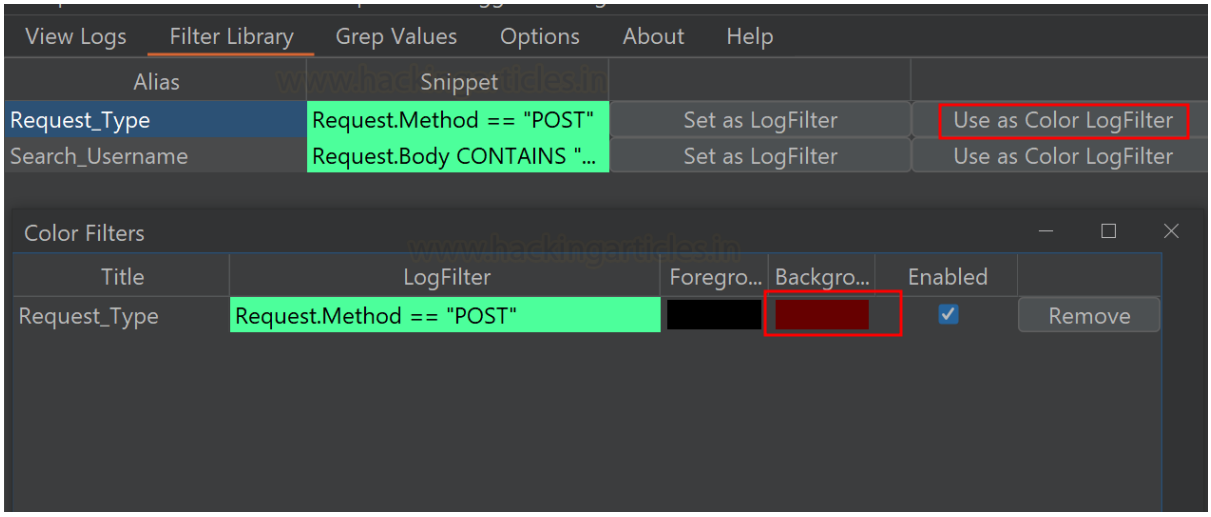
As you can see below, I have added a filter for

Request_Type: Request.Method == "POST"

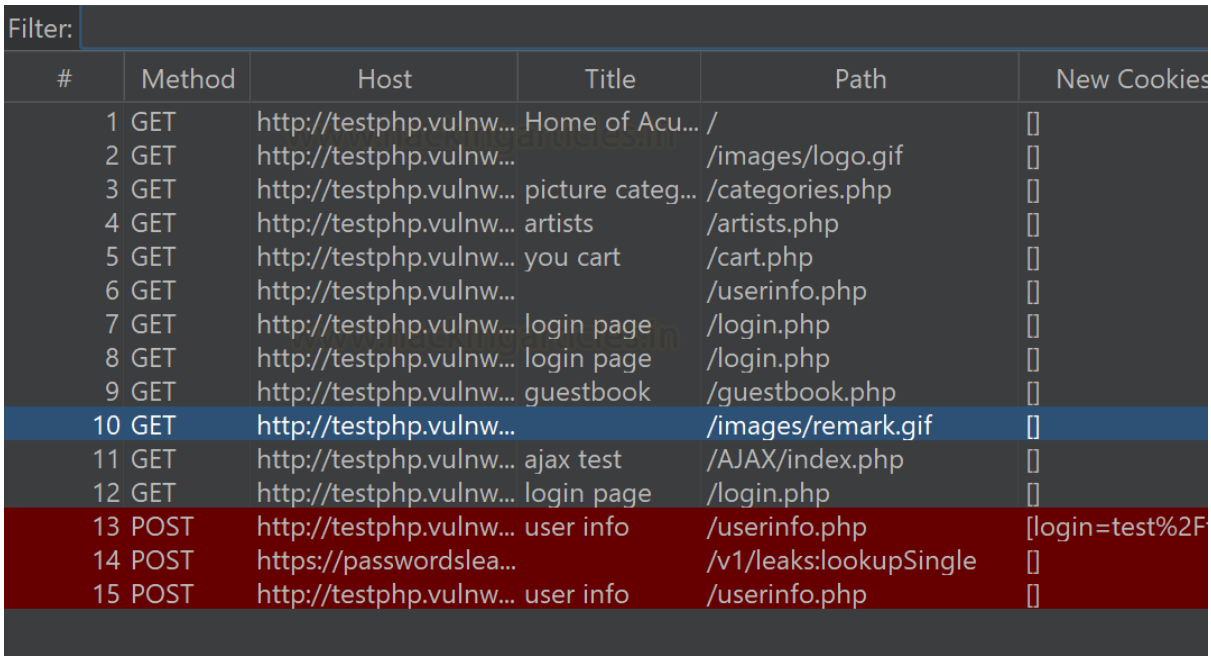


You no longer have to type repeatedly to find only POST requests. You can color-code this request so that the highlighted request stands out among all captured requests on the View Logs page.

Click on Use as **Color LogFilter** > **Select Background Color** > check Enable and save it.

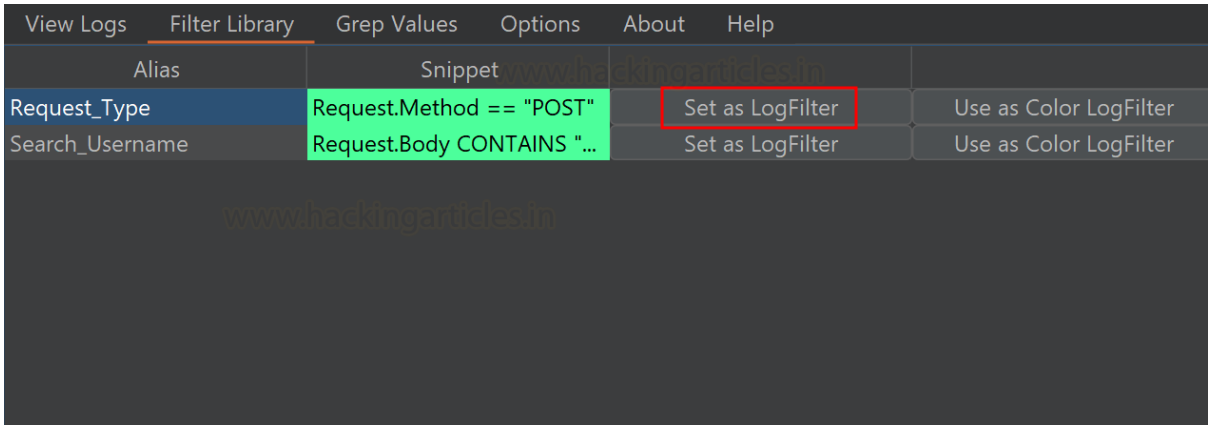


All POST requests are now highlighted in "Dard-Red" on this page.

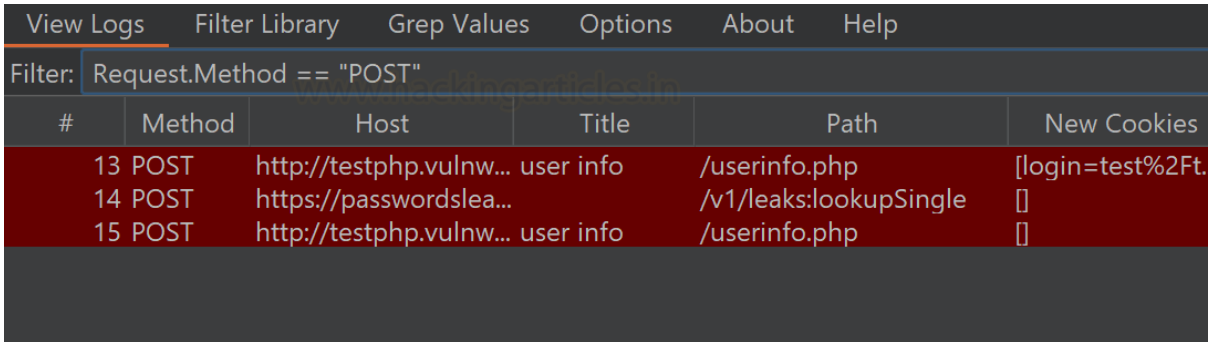


Similarly, you can save whole test scenarios in the Filter Library. There is two ways to call the saved filter:

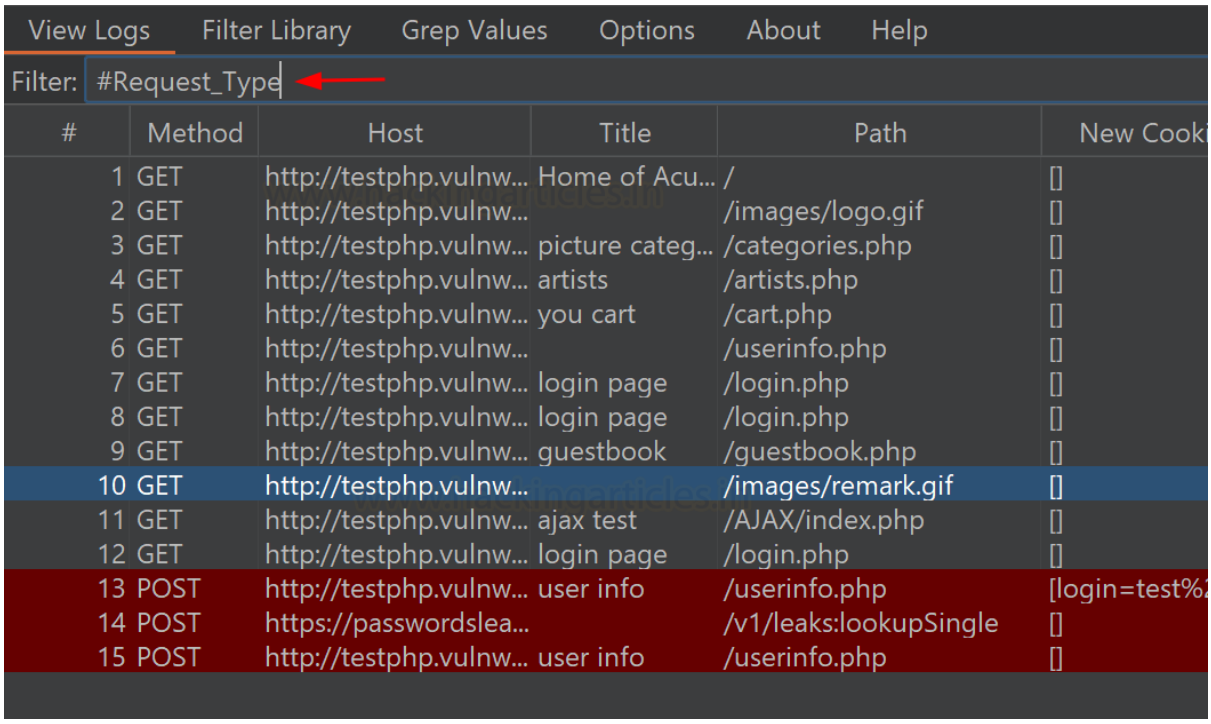
Method 1: In Filter Library, click on Set as LogFilter.



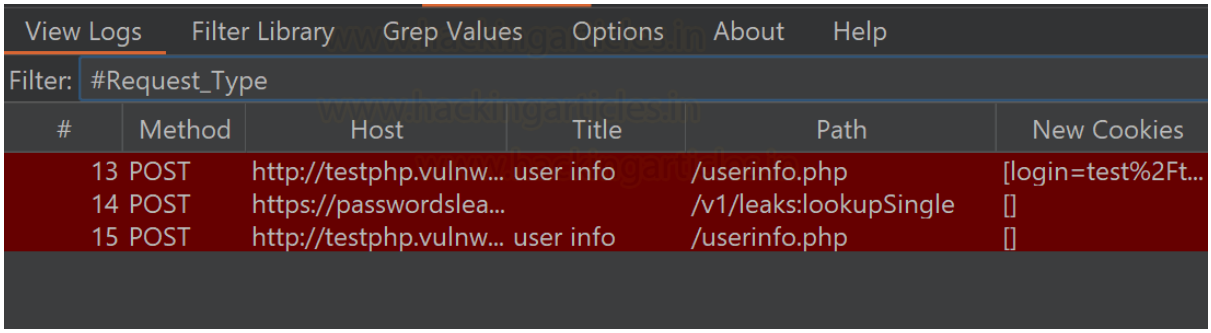
It will directly run the query and the desire result will be displayed.



Method 2: Use "#" with Alias name directly in filter bar.



And hit enter. The equivalent outcome will appear as follows:



Regex-Based Filter

Burp Logger's regex filter is a powerful feature that helps web security professionals pinpoint specific data within the vast sea of information during security testing.

You need to specify the regular expression (regex) pattern. This pattern acts like a search query, telling Burp Logger++ what kind of data you want to capture. You can create regex expression pattern to find data as like Email Address, IP Address, Server-side error messages, Software version disclosed, Any API Key exposed etc.

Go to Logger++, click on Grep Values tab. Here, you can see more filters to limit the search criteria.

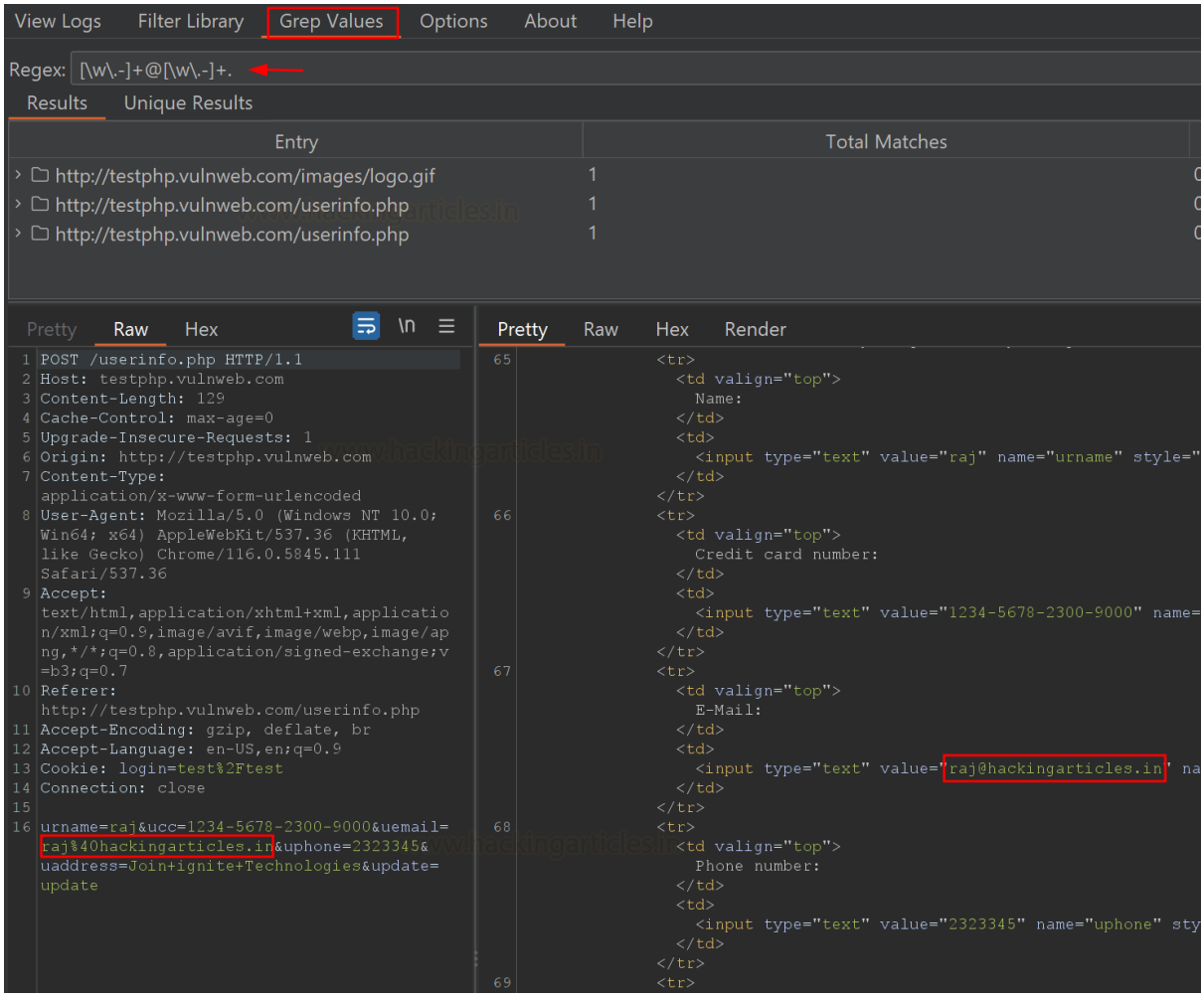
- Search Response = It will perform search only in responses.
- Search Request = It will perform search only in requests.
- In Scope Only = If you added the target URL in Scope only then it will only search within the scoped target.

For the time being, choose to search through every request and response. Let's take an example, if you want to find email addresses in web traffic, your regex pattern might look like

Regex:

[\w\.-]+@[\w\.-]+.

Directly paste this expression under Regex bar and press enter.



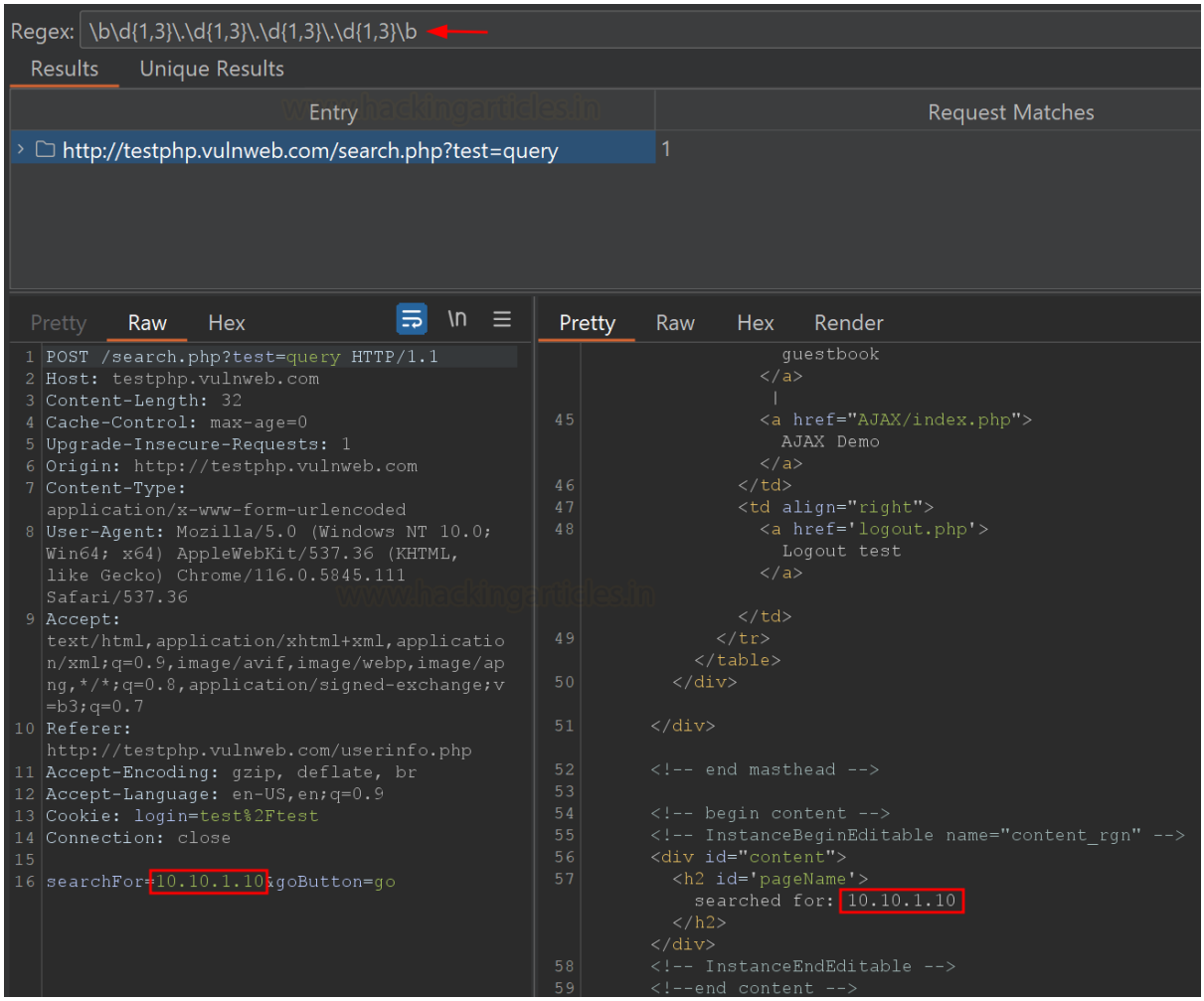
Consequently, the /userinfo.php request — which includes the email mentioned above is displayed.

You have two ways: Manually search through the complete request/response or click on Unique Result. The results that match the regex expression will be displayed only in Unique Results.

Similarly, Let's check for IP Addresses,

Regex Exp:

\b\d{1,3}\.\d{1,3}\.\d{1,3}\.\d{1,3}\b



It is evident that a POST request is being sent through the IP address 10.10.1.10.

In the same way, you can check for other important information like if you want to find the web traffic contains any FTP, HTTP, WWW.

Regex:

\b(ftp|www|http)[^\s]+

Export Data Feature

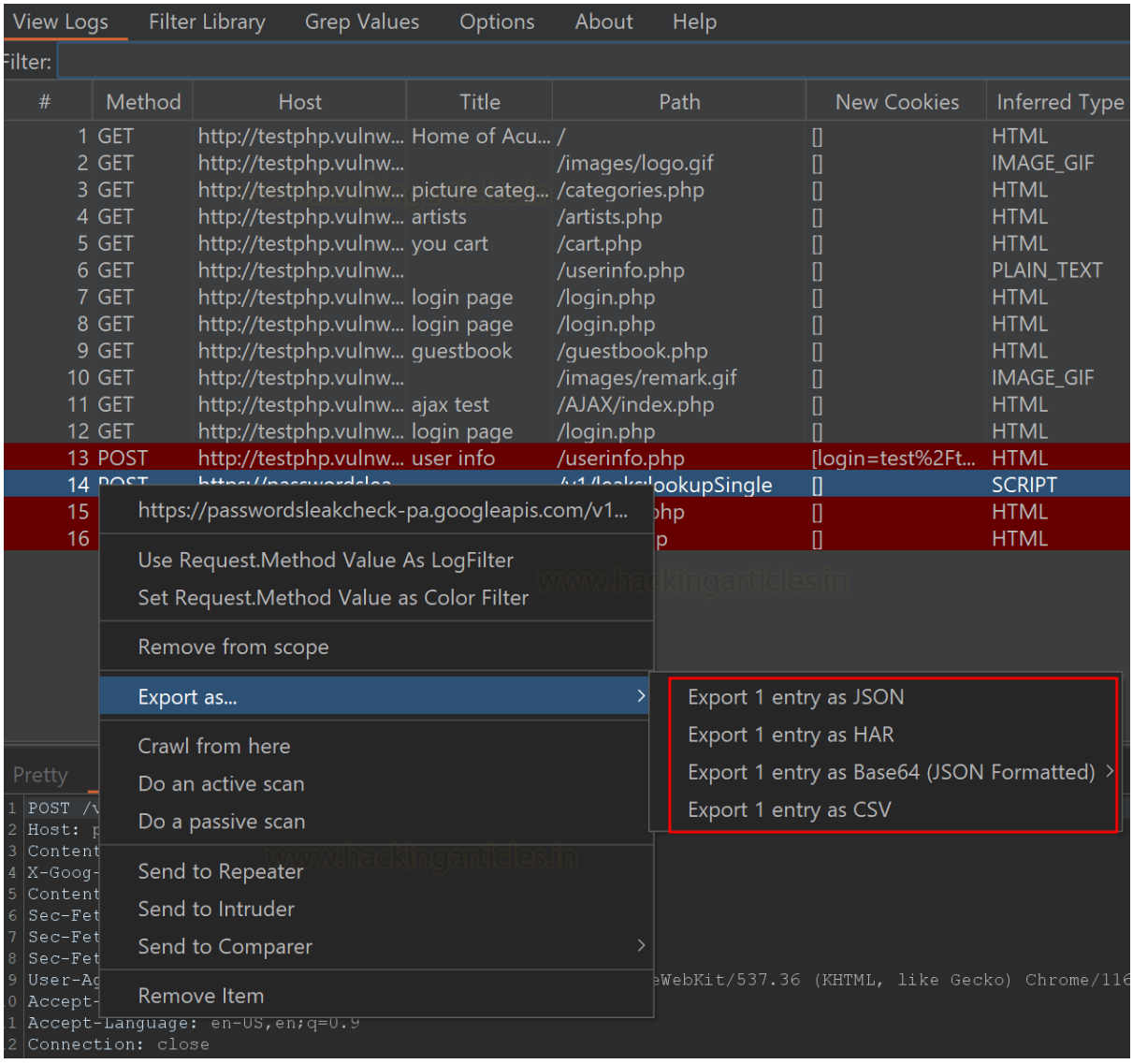
Burp Logger's data export feature is a valuable tool for web security professionals. It allows you to save, analyze, and share the captured data efficiently, making it an essential tool for documenting findings, performing in-depth analysis, and collaborating with others in the field of web security.

Why Export Data Feature is Helpful?

- **Data Preservation**: Exporting data from the Logger++ allows you to save a record of your testing session. This is essential for documentation and analysis.
- **External Analysis**: By exporting data, you can use external tools or software to perform in-depth analysis, generate reports, or share findings with team members.
- **Archiving Evidence**: It helps in preserving evidence of potential vulnerabilities or security issues discovered during testing, which is crucial for audits and compliance.
- **Collaboration**: Exported data can be easily shared with colleagues or experts for collaborative analysis, making it an asset in team-based security testing.
- **Customization**: Depending on the export format chosen, you can tailor the exported data to meet specific reporting or analysis requirements.

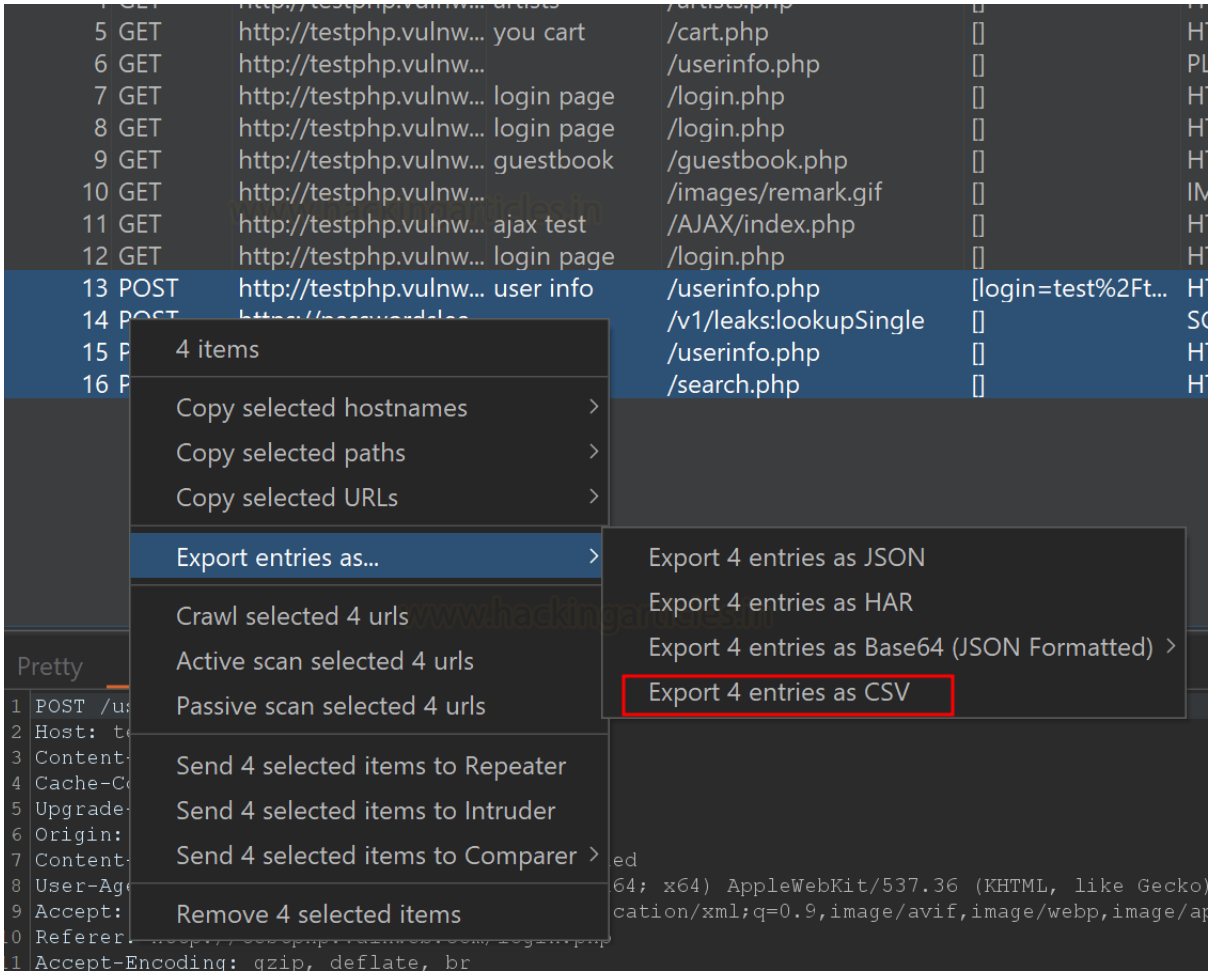
Supported Formats:

- **Base64 JSON Format**: Base64-encoded data is often used to include binary data within a JSON structure.
- **JSON Format**: JSON is a lightweight data-interchange format used for structured data.
- **CSV Format**: CSV files are widely supported and can be opened in spreadsheet software like Microsoft Excel or Google Sheets.
- **HAR Format**: HTTP Archive (HAR) format is used for capturing and storing the performance-related data. The HAR format contains detailed information about HTTP requests and responses.

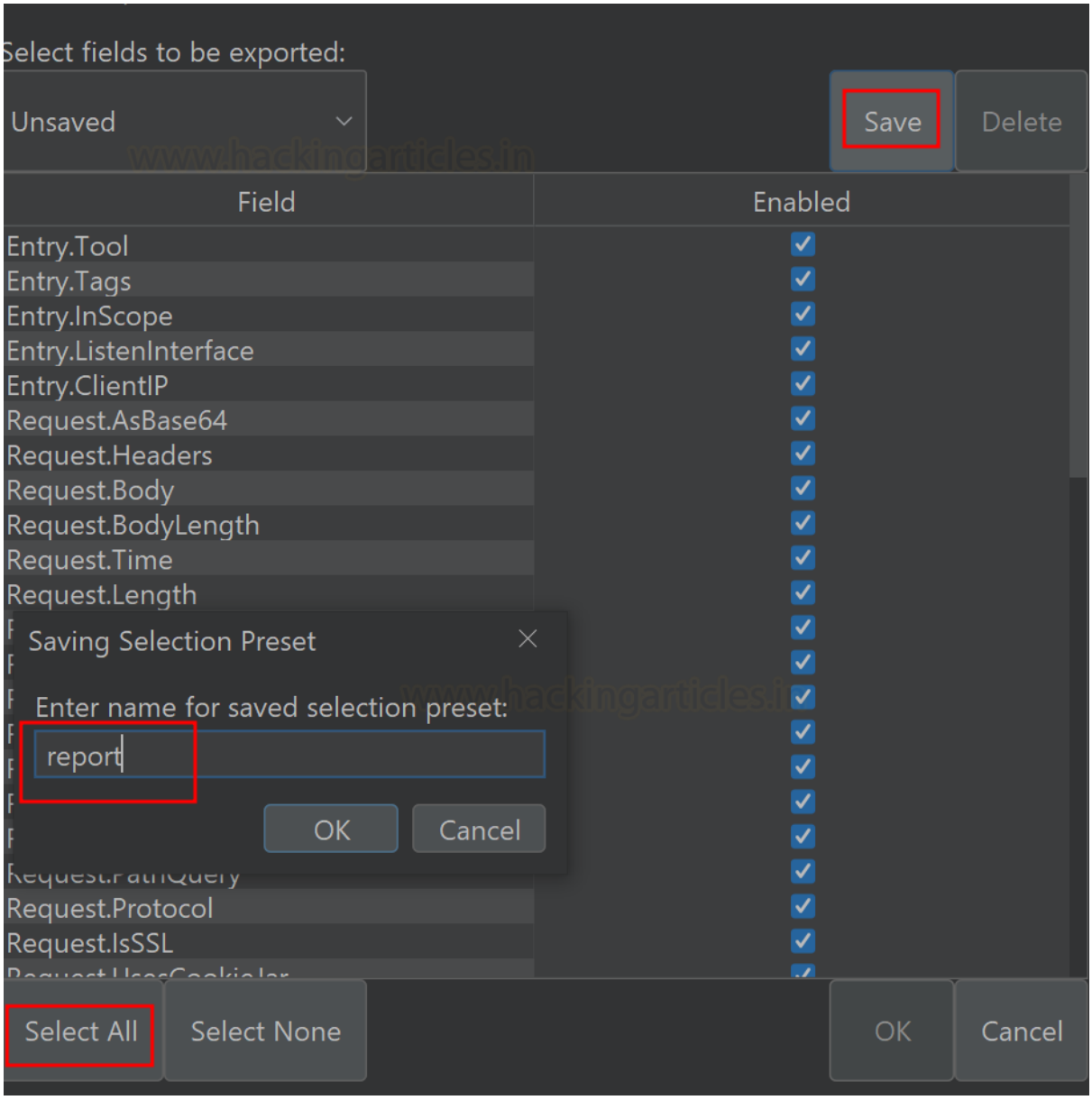


For Example, suppose you want to export all POST requests for further analysis.

Select the associate requests > **right click** > **choose Export entries as** > **Export as CSV**

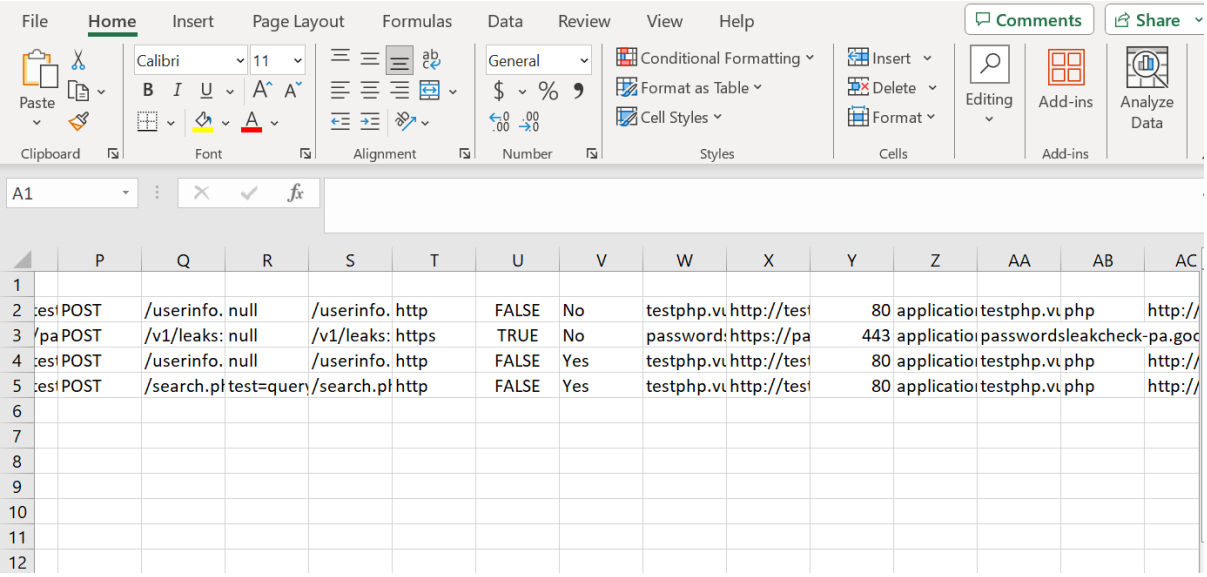


Now **Select All** > **Choose Save** > Enter the name and click on Ok.



Save the result to your system offline. You can examine the CSV file; it contains all of the values that you chose to save.

You may select the only required values to store based on your needs.



	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC
1														
2	test	POST	/userinfo.	null	/userinfo.	http	FALSE	No	testphp.vu	http://test	80	application	testphp.vu	php	http://
3	/pa	POST	/v1/leaks:	null	/v1/leaks:	https	TRUE	No	password	https://pa	443	application	passwordsleakcheck-pa.goo	
4	test	POST	/userinfo.	null	/userinfo.	http	FALSE	Yes	testphp.vu	http://test	80	application	testphp.vu	php	http://
5	test	POST	/search.ph	test=query	/search.ph	http	FALSE	Yes	testphp.vu	http://test	80	application	testphp.vu	php	http://
6														
7														
8														
9														
10														
11														
12														

Conclusion

Whether you're doing thorough security checks or searching for hidden issues on your web applications, Burp Suite gives you a set of extensions to help you as a penetration tester. These tools boost the efficiency and effectiveness of your security assessments on web applications.

Hence, one can make use of these commands as a cybersecurity professional to assess vulnerabilities on systems and keep these systems away from threat.

References

- https://www.hackingarticles.in/burpsuite-for-pentester-autorize/
- https://www.hackingarticles.in/burpsuite-for-pentester-logger/
- https://www.jython.org/download.html
- https://github.com/lwierzbicki/RegexFinder/blob/main/burp.regex.tsv